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Photonics-integrated power electronics: a review and perspective

Xin Yang^{1,2*}, Zineng Yang^{1,2}, Jizhong Jiang^{1,2}, Hengbo Zhang^{1,2}, Feng Zhou³, Shailesh N. Joshi³, Ercan M. Dede³, Liyang Jin⁴, Chao Xiang², Han Wang^{1,2}, Linbo Shao^{4,5}, and Yuhao Zhang^{1,2*}

¹Centre for Advanced Semiconductors and Integrated Circuits (CASIC), The University of Hong Kong, Hong Kong SAR, People's Republic of China

²Department of Electrical and Computer Engineering, The University of Hong Kong, Hong Kong SAR, People's Republic of China

³Electronics Research Department, Toyota Research Institute of North America, Ann Arbor, Michigan, United States of America

⁴Bradley Department of Electrical and Computer Engineering, Virginia Tech, Blacksburg, VA, United States of America

⁵Department of Physics and Center for Quantum Information Science and Engineering (VTQ), Virginia Tech, Blacksburg, VA, United States of America

*E-mail: xyang.hku@connect.hku.hk; yuhzhang@hku.hk

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The rapid advancement of wide-bandgap (WBG) and ultra-WBG power semiconductors has propelled modern power electronics toward high switching frequencies and power densities. However, these high-stress operating regimes introduce severe electromagnetic interference and galvanic isolation challenges that traditional electrical control and monitoring methods struggle to mitigate. To overcome these fundamental physical bottlenecks, the integration of photonic technologies into power electronic systems has emerged as a promising solution. By utilizing photons rather than electrons as the primary medium for signal, power, and data transfer, optical components provide reliable galvanic isolation and near-zero parasitic coupling capacitance. This article presents a comprehensive review and perspective on the state-of-the-art in optical power electronics. We systematically examine the underlying physics, structural innovations, and hardware demonstrations of key optical components for power applications, including advanced photodetectors, optically-controlled power semiconductor switches, optocoupler-based signal isolators, photovoltaic-based auxiliary power supplies, monolithically integrated optical driver integrated circuits, optical sensors for voltage, current, and temperature monitoring, and the emerging applications in quantum sensing and photonic systems. By benchmarking current technological capabilities and outlining future roadmaps, this review provides a reference for developing the next generation of resilient, ultra-compact, and intelligent optical power electronics. © 2026 The Author(s). Published on behalf of The Japan Society of Applied Physics by IOP Publishing Ltd

1. Introduction

Wide-bandgap (WBG) power devices, such as silicon carbide (SiC) metal-oxide-semiconductor field-effect transistor (MOSFET) and gallium nitride (GaN) high electron mobility transistors (HEMTs), have significantly advanced the field of power electronics by enabling high efficiency, switching frequencies, and power densities.^{1–4)} The continuous evolution of these devices is actively driving modern applications, including electric vehicles, artificial intelligence (AI) data centers, photovoltaic (PV) renewable energy systems, and high-energy pulsed power applications,^{5,6)} toward ever-higher switching frequencies,^{7–11)} increased power ratings,^{12–17)} and greater system compactness.^{18–22)} Beyond traditional power conversion, WBG devices are now recognized as critical components for achieving global carbon neutrality goals.²³⁾ Furthermore, their unique material properties are opening doors to highly unconventional applications within the power electronics society, such as the recent proposal of a SiC MOSFET-based cryogenic neuromorphic network.²⁴⁾ Pushing these boundaries even further, WBG and ultra-WBG (UWBG) power modules have recently demonstrated medium-voltage (MV), megawatt switching capabilities, paving the way for next-generation, extreme-power applications.²⁵⁾

As an example of critical power applications, Fig. 1 shows the 800 VDC power supply architecture for AI data centers, which comprises essential power electronics units such as AC–DC and DC–DC conversion modules. Driven by the goal of directly converting high-voltage AC power from the grid (e.g. 10 kV) to a 48 V DC bus, these systems demand increasingly high switching frequencies and power densities.

However, pushing these performance boundaries introduces severe operational challenges, primarily centering around strict galvanic isolation requirements and electromagnetic interference (EMI).^{26–30)} First, modern digital control systems and low-voltage sensing circuits within power electronics are highly sensitive to EMI, which can induce false triggering, logic errors, or severe control failures.³¹⁾ Second, the rapid dv/dt and di/dt switching transients characteristic of fast-switching WBG devices act as intrinsic EMI sources, generating high-frequency common-mode noise that propagates throughout the system.³²⁾ Third, to ensure safe and reliable operation, the gate driving circuits should maintain reliable galvanic isolation from the high-voltage power stage to prevent high-voltage breakdown and protect the low-voltage control domain.^{33,34)} Fourth, the gate driving circuits for high-side switching devices in bridge topologies require an independent, highly isolated auxiliary power supply (APS) to maintain the necessary gate-to-source driving voltage.²⁷⁾

To address these critical challenges, the integration of optical components into power electronics systems provides a promising solution.^{35–37)} The primary advantage of utilizing light is the intrinsic EMI-free characteristics of optical signals: as the information carrier is a photon rather than an electron, optical transmission is resistant to electromagnetic noise. Furthermore, translating electrical signals into the optical domain effective galvanic isolation, decoupling the sensitive control systems from the noisy power loops.

The fundamental differences between electrical and optical transmission methods are illustrated in Fig. 2. The carrier for electrical signal transmission [Fig. 2(a)] is typically composed of copper power cables or printed circuit



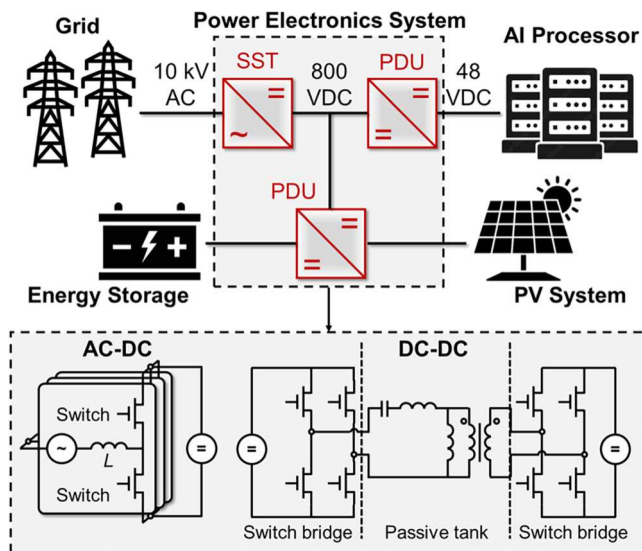


Fig. 1. System schematics of 800 V DC power delivery architecture in a data center, including solid-state transformers (SST) and power delivery units (PDU). Two typical circuit topologies for AC-DC and DC-DC power conversion are also shown there.

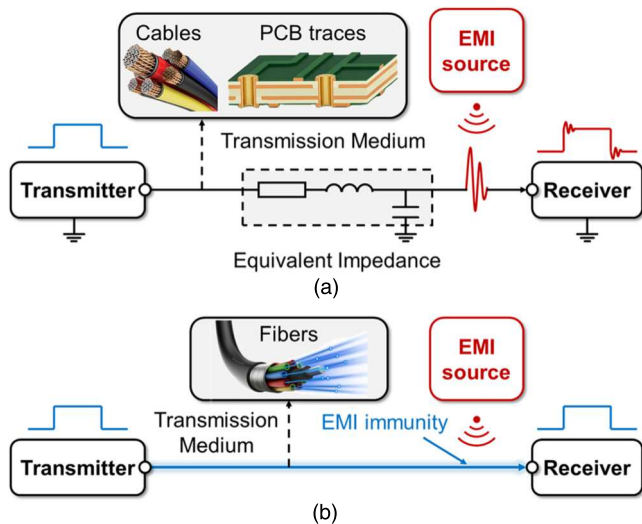


Fig. 2. System schematics and the impact of EMI in (a) electrical transmission and (b) optical transmission.

board (PCB) traces. Physically, these conductors act as an equivalent RLC circuit possessing parasitic resistance, inductance, and capacitance. During high-frequency operation, this equivalent impedance distorts the transmitted signals and serves as both an antenna for radiating EMI and a receiver that absorbs it. Conversely, as shown in Fig. 2(b), the signal transmission in an optical system utilizes optical fibers or transparent insulating media. This optical approach allows for long-distance transmission with negligible propagation delay and EMI susceptibility, ensuring the received control signals remain intact regardless of the surrounding electro-magnetic noise.

Motivated by these advantages, research efforts have recently been dedicated to developing and integrating various optical components directly into power electronics systems. Figure 3 summarizes the primary functions and applications of photonic components within this domain, categorized into five main areas: (1) optically-controlled power semiconductor switches, (2) signal isolation, (3)

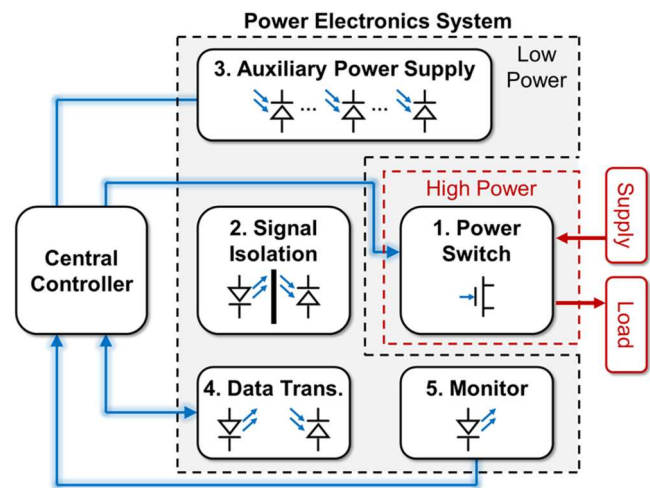


Fig. 3. Functions of photonic components in power electronics systems.

APSS, (4) data transmission, and (5) system monitors and sensors.

This article reviews these topics comprehensively, discussing their basic working principles, structural optimizations, performance comparisons with traditional electrical designs, and the underlying physical mechanisms, while providing strategic prospects for the future development of optical power electronics. Compared with the earlier overview on this topic published a decade ago,³⁸⁾ which primarily focused on optically triggered semiconductor switches, this article aims to advance the field from two distinct perspectives. First, it incorporates the latest technological breakthroughs achieved over the past decade, particularly highlighting several state-of-the-art optical triggered power devices for ultrafast switching. Second, it adopts a highly systematic, system-level approach by covering a much broader spectrum of optical power components as summarized above.

The remainder of this paper is organized as follows: chapter 2 introduces the performance metrics of photodetectors based on different semiconductor materials, providing a foundational physical analysis of key parameters that are critical for target power electronics applications. Chapter 3 provides an overview of optically-controlled power semiconductor devices, detailing their schematics and working principles. It further highlights the potential limitations of the optical gate driving approach and discusses corresponding solutions. Chapter 4 discusses typical techniques for optical signal isolation in power electronics, comparing key parameters, long-term reliability, and dynamic performance at high and cryogenic temperatures. Chapter 5 explores the working principles of PV cell-based APSS, comparing different PV materials and benchmarking the efficiency of photonic power supplies against traditional air-core and magnetic-core transformers. Chapter 6 examines system schematics relying on optical fibers for high-speed data transmission and protection, including combined power-and-data delivery and the monolithic integration of power ICs with optical receivers. Chapter 7 reviews the state-of-the-art in optical sensors for voltage, current, and temperature measurement. Chapter 8 explores the emerging applications in quantum sensing and photonic systems. Finally,

chapter 9 concludes the paper and offers insights into the future trajectory of optical power electronics.

2. Photodetectors based on varying materials

As the basic components to achieve optical-electrical conversion, the photodetector serves as the foundational transducer for virtually all subsequent optical components discussed in this review. The suitability of a photodetector for these diverse applications is primarily evaluated through two critical performance metrics: the maximum responsivity ($R_{\max}$) and the highest operating bandwidth ($f_{3\text{dB}}$). Responsivity quantifies photoelectric conversion efficiency and is defined as:

$$R_{\max} = \eta q / h\nu \quad (1)$$

where η is the external quantum efficiency, q is the charge of single electron, and $h\nu$ is the energy of single incident photon.³⁹⁾

Conversely, the operating bandwidth dictates the transient tracking capability and is fundamentally limited by two factors: the carrier transit-time bandwidth limit (f_{tr}) and the parasitic resistor-capacitor (RC) bandwidth limit (f_{RC}). Following the established model for high-frequency photodetectors,⁴⁰⁾ the overall 3 dB bandwidth is conventionally approximated by:

$$f_{3\text{dB}} \approx (f_{\text{tr}}^{-2} + f_{\text{RC}}^{-2})^{-0.5}. \quad (2)$$

The RC-limited bandwidth is given by:

$$f_{\text{RC}} = \frac{1}{2\pi R_{\text{tot}} C} \quad (3)$$

where C is the junction capacitance, which is inversely proportional to the depletion width (W), and R_{tot} is the equivalent total resistance. The f_{tr} is determined by the average carrier saturation velocity (v_{sat}) and is typically expressed as:⁴⁰⁾

$$f_{\text{tr}} \approx \frac{0.44 v_{\text{sat}}}{W}. \quad (4)$$

Because f_{tr} is inversely proportional to W (due to longer carrier transit distances) while f_{RC} is directly proportional to W (due to the reduction in junction capacitance), the overall bandwidth is typically limited by just one of these two factors. Therefore, device designers must carefully optimize the depletion width depending on the dominant limiting factor of the target operation regime, which in turn limits the thickness of the absorption region and bounds the overall device responsivity.^{40,41)}

Among commercially established technologies, elemental semiconductors such as silicon (Si) and germanium (Ge) demonstrate robust performance from the ultraviolet (UV) to the mid infrared regimes. Si provides a mature, highly integrated platform for visible to near-infrared (NIR) detection with responsivities reach 0.72 A W^{-1} and bandwidths of 40 GHz, making it a ubiquitous choice for standard industrial opto-isolation.^{41–43)} For higher responsivity, avalanche photodiode (APD) can reach more than 50 A W^{-1} ⁴⁴⁾ and highly engineered ones can reach single photon sensitivity, known as single photon avalanche diode (SPAD). However, APDs and SPADs need large bias voltages close to the

breakdown voltage, and thus they are more vulnerable to EMI. For higher-speed data transmission at telecom wavelengths, Ge and III–V group compounds, such as gallium arsenide (GaAs), indium phosphide (InP), and indium gallium arsenide (InGaAs), are utilized. Ge-on-Si photodiodes (PDs), for instance, have achieved high optoelectrical bandwidths up to 265 GHz by leveraging a germanium fin structure sandwiched between complementary in situ doped silicon layers that optimize the electric field distribution for rapid carrier extraction.⁴⁵⁾ Similarly, InGaAs and InP photodetectors utilize various architectures depending on specific speed and efficiency requirements. Uni-traveling-carrier architectures decouple the slow hole mobility from the transit time, enabling bandwidths exceeding 170 GHz.⁴⁶⁾ Concurrently, p-i-n PD architectures are extensively utilized; for example, high-efficiency mushroom-mesa waveguide p-i-n structures⁴⁷⁾ and monolithically integrated InGaAs nanostructure p-i-n photodetectors⁴⁸⁾ demonstrate excellent high-speed optical data reception. These ultra-high-bandwidth characteristics are strictly required when translating gigahertz-level optical control signals to drive modern WBG switching devices without introducing detrimental propagation delays.

While narrow-bandgap materials excel in ambient, high-speed telecommunication environments, the severe thermal and electromagnetic conditions inherent to modern high-power modules necessitate materials with extreme environmental robustness. The thermally generated dark current density (J_{dark}) in a semiconductor scale can be expressed as:³⁹⁾

$$J_{\text{dark}} \propto \exp\left(-\frac{E_g}{2kT}\right) \quad (5)$$

where E_g is the bandgap and k is the Boltzmann constant.

Consequently, WBG and UWBG semiconductors, such as GaN (3.4 eV), SiC (3.3 eV), and aluminum nitride (AlN) (6.2 eV), inherently suppress thermal noise. Their continuous characterization and reliability evaluation under extended industrial and extreme temperature grades remain a critical focus for advancing safe power module integration.^{49,50)} Furthermore, these materials natively absorb in the UV spectrum, providing a “solar-blind” characteristic that renders the optical receiver virtually immune to ambient visible and infrared interference inside an open converter cabinet.

Recent advancements in the synthesis of high-quality single-crystalline SiC thin films have further enabled highly flat, miniaturized deep-UV detectors that exhibit highly stable baselines across wide temperature, bias, and humidity variations.⁵¹⁾ Beyond simple detection, SiC is emerging as a dominant complementary metal-oxide-semiconductor (CMOS)-compatible integrated photonics platform; it inherently exhibits the Pockels effect and can sustain stable operation at extreme optical intensities ($>900 \text{ kW mm}^{-2}$), paving the way for monolithic integration of optical sensors and receivers.⁵²⁾ Similarly, GaN devices consistently deliver responsivities in the tens of A/W alongside sub-nanosecond response times, positioning them as ideal candidates for monolithic integration with GaN HEMTs.^{53–55)}

Looking beyond conventional bulk crystals, emerging low-dimensional materials and metal halides perovskite offer structural and optoelectronic flexibility for next-generation

optical power electronics. Perovskite materials provide highly tunable, broad UV-NIR absorption with exceptionally high responsivity and rapid photo responses, achievable through low-cost solution processing.⁵⁶⁾ Simultaneously, two-dimensional (2D) materials are demonstrating significant potential for ultra-high-speed and high-sensitivity optical switching. Graphene, benefiting from its ultra-high carrier mobility, provides an ultra-broadband visible-to-far-infrared detection range with bandwidths surpassing 500 GHz.^{57–59)} Transition metal dichalcogenides, such as MoTe₂, circumvent the zero-bandgap limitation of graphene, exhibiting high responsivity values as high as 2560 A W⁻¹ due to strong light-matter interactions and photogating effects.⁶⁰⁾ Moreover, the lack of out-of-plane dangling bonds in these 2D materials eliminates the strict lattice-matching constraints required in traditional epitaxy. This distinct advantage of van der Waals heterointegration offers a novel pathway for directly fabricating vertically integrated, optically-driven photogate circuits directly atop mature Si, SiC, or GaN power device platforms.

A comprehensive benchmark of the performance metrics of these devices is summarized in Table I, detailing the detection wavelengths, $R_{\max}$, peak bandwidth (BW_{Peak}), and operational temperature ranges across elemental, compound, metal halide, and low-dimensional semiconductor platforms.^{40–43,45–85)} Notably, most of these materials support heterogeneous or monolithic integration with WBG and UWBG devices, facilitating the corresponding photodetector implementation alongside power devices.

3. Optical power semiconductor devices

Power semiconductor devices serve as the essential building blocks for managing high-power conversion in power converters. In standard switching applications, they operate by switching at high frequencies between a high current conduction state with small voltage drop, and an off-state state capable of sustaining high blocking voltages. The fundamental performance metrics for these devices are the reduction of conduction and switching losses alongside the pursuit of enhanced switching speeds.

To reduce the parasitic inductances, propagation delays, and significant EMI susceptibility inherent in conventional electrical gate drivers, recent research focuses on directly integrating optical receiving elements with the power semiconductor switch itself. This bypasses intermediate electrical signal conversion, translating optical light signals directly into switching actions. Broadly speaking, optically controlled power devices are designed using one of two fundamental approaches: (a) leveraging direct photogeneration combined with high bipolar avalanche gain to sustain the device's primary conduction current, or (b) utilizing low-power optical signals to control or drive the device's gate or base, thereby modulating the main conduction current.

3.1. Switches based on direct photogeneration

A representative example of the first approach is the photoconductive semiconductor switch (PCSS). The device schematic and ideal switching waveforms of a PCSS are shown in Fig. 4, respectively.^{86,87)} Operating in bulk material without complex junction structures, the PCSS is triggered when a high-intensity pulsed laser illuminates the semiconductor layer, generating large quantities of electron-hole

pairs. To ensure a high bipolar gain, the PCSS operates under a high DC voltage that makes the device approach the avalanche regime. Upon illumination, the produced carriers initiate rapid impact ionization and avalanche multiplication under large electric field. Because the carrier generation occurs near-instantaneously throughout the bulk volume, PCSS devices exhibit virtually zero jitter and sub-nanosecond rise times, making them well-suited for pulsed power applications, such as particle accelerators, ground-penetrating radar, and munition firing sets.⁸⁷⁾

WBG materials like SiC and GaN are attractive for PCSS due to their large critical electric field strengths (3.0–3.5 MV cm⁻¹) and deep extrinsic compensating levels (e.g. vanadium in SiC) that allow for high-voltage blocking in the dark state.^{37,88)} However, the fundamental limitation of the PCSS is its reliance on high DC bias to sustain impact ionization and the high optical peak power required from the triggering laser, restricting its use to specialized pulsed applications rather than regular power switching applications.

For high-power switching applications, the light-triggered thyristor (LTT) relies on photogenerated electron-hole pairs combined with inherent high bipolar regenerative gain to sustain high device currents. As shown in the schematic and operational diagram in Fig. 5, an LTT features a specialized central optical window where incident photons generate a primary photocurrent. This initial current acts as the base drive for the internal p-n-p-n regenerative structure, rapidly latching the device into full conduction.³⁷⁾ Modern silicon-based LTTs are the promising switching devices in ultra-high-power systems, specifically in high-voltage DC transmission, where individual wafer-scale devices can block up to 8.5 kV and conduct 6 kA.⁸⁹⁾ Advanced LTT designs also monolithically integrate crucial overvoltage and dv/dt protection functions directly into the silicon die.⁹⁰⁾

To push the performance limits of LTTs, researchers have demonstrated SiC-based LTTs and pulsed-power transistors. Zhao and Islam demonstrated a 1200 V SiC transistor triggered by a 0.5 mJ UV laser, achieving a 20 ns rise time and 200 ns fall time.⁹¹⁾ Similarly, Wang et al. developed a 4H-SiC thyristor featuring a thin double-layer n-base structure to enhance hole injection, allowing the device to be triggered by a continuous UV light-emitting diode (LED) with an intensity of just 100 mW cm⁻².⁹²⁾ Despite their power-handling capabilities, LTTs are hampered by an inherent trade-off between the required optical trigger power and the allowable switching speed, as higher bipolar gain could reduce the trigger power but slow down the device switching. In addition, LTTs usually exhibit long propagation delays and could not be actively turned off via light.

3.2. Switches based on optical gate control

The second major approach for developing optically-controlled power switches relies on photogenerated electron-hole pairs to modulate the conductivity in a gated channel, with the aim to decouple the optical trigger power from the main conduction current. Palmese et al. demonstrated an optically-gated AlInN/GaN HEMT, where a 375 nm laser depleted the two-dimensional electron gas barrier, achieving sub-microsecond switching times.⁹³⁾ To maximize the light-matter interaction volume, Hsia et al. pioneered a series of optically-controlled FinFETs. Developed from their first generation of electrically-driven Fin-MOSFET^{94,95)} and

Table I. Comparison of key parameters for photodetectors based on various semiconductor materials.

Group	Semiconductor	Detection wavelength	^{a)} R_{max} (A W ⁻¹)	^{b)} BW_{Peak} (GHz)	Operating temp
Elemental semiconductor	Si	UV-NIR (Intrinsic absorption) and SWIR (mid-bandgap absorption)	50 ⁴²⁾	40 ⁴¹⁾	Extended Industrial Grade
III–V group compound	Ge	NIR-SWIR	16.8 ⁵⁹⁾	265 ⁵⁸⁾	Cryogenic Grade/Room Temperature ⁶²⁾
	GaAs	VIS-SWIR	0.7 ⁶⁹⁾	54 ⁶⁹⁾	Cryogenic Grade/Room Temperature ⁶⁹⁾
	GaN	UV	35.4 ⁵³⁾	0.9 ⁵⁴⁾	Extended Industrial Grade/High-Temperature Grade ⁵³⁾
	InP	VIS-NIR	3000 ⁷⁹⁾	170 ⁴⁶⁾	Cryogenic Grade to Room Temperature
	AlN	UV	0.39 ⁵⁰⁾	3.5 ⁴⁹⁾	Room Temperature/High-Temperature Grade ⁸⁴⁾
II–VI group compounds	CdZnTe	VIS	0.5 ⁷²⁾	0.35 ⁷²⁾	Room Temperature
	HgCdTe	SWIR-LWIR	4.98 ⁷⁶⁾	104 ⁷⁶⁾	Cryogenic Grade ⁶⁷⁾
IV–IV group compounds	SiC	UV	4×10^5 ⁸³⁾	7.1 ⁵²⁾	Extended Industrial Grade/High-Temperature Grade ⁵¹⁾
Multi-component compounds	InGaAs	SWIR	6.5×10^3 ⁸⁵⁾	25 ⁴⁸⁾	Cryogenic Grade ⁶³⁾ /Room Temperature
Metal halides	Perovskite	UV-NIR	139 (Cs ₃ Bi ₂ Br ₉) ⁸⁰⁾	0.0257 (FAPbI ₃) ⁸¹⁾	Extended Industrial Grade ⁸²⁾
Low-dimensional materials	Graphene	VIS-FIR	500 ⁵⁸⁾	500 ⁵⁹⁾	Cryogenic Grade/Room Temperature ⁵⁷⁾
	Transition metal dichalcogenides (TMDs)	UV-SWIR	2560 for MoTe ₂ ⁶⁰⁾ /880 for MoS ₂ ⁷⁸⁾	300 ⁷⁷⁾	Extended Industrial Grade

a) R_{max} values represent the highest reported responsivity, which may include internal gain mechanisms to showcase the material’s sensitivity limit. b) BW_{peak} values denote the state-of-the-art 3 dB bandwidth achieved based on various materials.

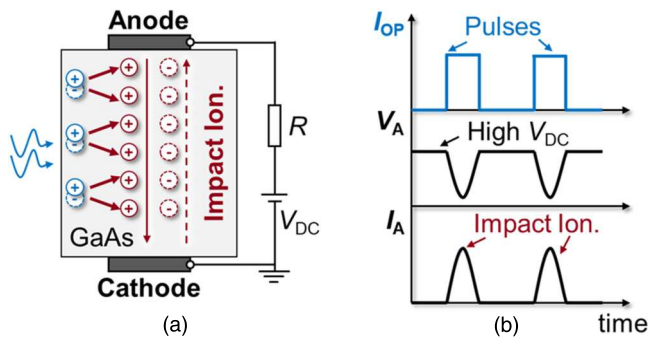


Fig. 4. (a) Device schematic and (b) ideal switching waveforms of a PCSS. The blue symbols represent photogenerated electron-hole pairs, while the red symbols represent carriers generated by impact ionization.

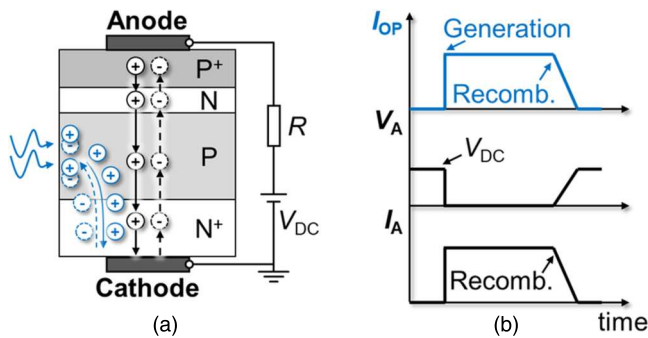


Fig. 5. (a) Device schematic and (b) ideal switching waveforms of an LTT. The blue symbols represent photogenerated carriers, while the black symbols represent the carriers supporting the main device conduction current.

second generation of electrically-driven fin-junction field-effect transistor (JFETs),^{8,96,97} the schematic and working principles and device schematic of this optically-controlled FinFETs are shown in Fig. 6. By illuminating the depleted fin channel with a 365 nm UV LED, massive photogeneration turns the device on.⁹⁸ Over a developmental cycle, this architecture demonstrates an on-off current ratio of $>10^7$, an optical gain of 10^4 A W^{-1} , and an off-state breakdown exceeding 600 V.⁹⁹ Subsequent gate stack engineering that transitions from the MOS gate into a junction gate allowed for optical triggering at longer, more accessible wavelengths ($\geq 395 \text{ nm}$),¹⁰⁰ and the integration of a p-NiOx/n-GaN heterojunction reduced minority carrier screening, enabling kHz-level operation with a turn-off time reduced to 90 μs .¹⁰¹

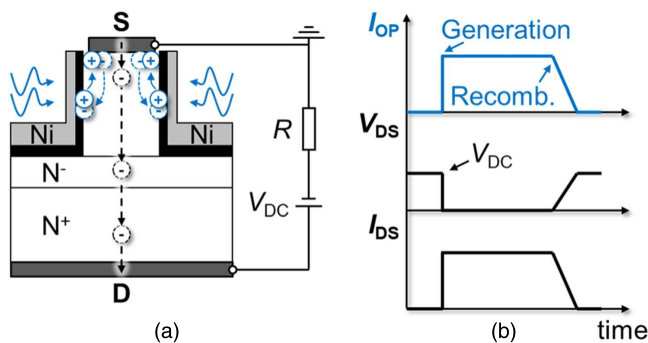


Fig. 6. (a) Device schematic and (b) ideal switching waveforms of an optical-gating GaN FinFET. The blue symbols represent photogenerated carriers, while the black symbols represent the carriers supporting the main device conduction current.

However, a critical challenge facing optically-gated power switches is the lack of a robust discharge path for stored charges during high-voltage dynamic switching. In typical power electronics applications, merely removing the turn-on signal is insufficient; the system must also provide an effective, low-impedance path to rapidly discharge the gate, thereby facilitating high switching frequencies and minimizing switching losses. As shown in Fig. 7(a), a conventional electrical driver (e.g. a CMOS IC) readily provides this low-impedance path to actively charge and discharge the device's gate. In contrast, for direct optical gating devices [Fig. 7(b)], while the turn-on charging current is supplied by the applied optical power, the turn-off process relies entirely on slow intrinsic carrier recombination, resulting in prolonged turn-off times. This discharging delay is often exacerbated by the persistent photoconductivity effect commonly observed in WBG semiconductors.^{102,103} Furthermore, once the light source is removed, the gate is effectively left floating. Under high-voltage switching transients, this floating gate state can induce large voltage overshoots and hazardous parasitic oscillations.

To resolve the dynamic turn-off limitations, the hybrid optical gate scheme utilizes distinct optical components to produce active gate charging and discharging currents in the switching transient. Figure 8 depict the early simulated schematic and ideal waveforms of an optically-controlled MOSFET proposed by Sarkar and Mazumder.¹⁰⁴ In this concept, a PD is connected to the gate to modulate the channel conductivity for device turn-on, while an additional photoconductive switch is connected between the gate and source to provide a low-resistance discharge path during device turn-off. Moving from simulation to hardware, Mazumder et al. developed a generalized optical driving scheme utilizing two customized optically-triggered power transistors (OTPTs).^{105–107} As shown in Fig. 9, these OTPTs are low-voltage, bipolar GaAs transistors placed directly in the gate loop: one acts as a pull-up to inject photogenerated gate current, while the other acts as a pull-down to rapidly discharge the gate. This architecture was experimentally validated for the control of Si CoolMOS, 1200 V SiC MOSFETs, and 15 kV single-bias emitter turn-off thyristors.^{106,107} Despite the high switching performance, this approach requires the customized design and fabrication of bipolar OTPTs, as well as auxiliary electrical drivers to manage the dual optical signals, limiting its widespread adoption in industrial systems.

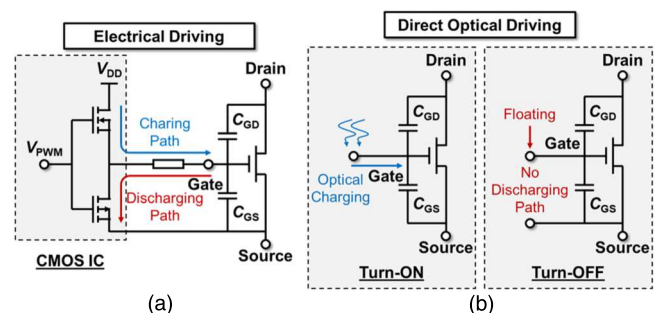


Fig. 7. Equivalent circuit schematic of (a) electrical driving and (b) direct optical gate driving in power device switching transient, highlighting the essential role of the discharging path for fast turn-off.

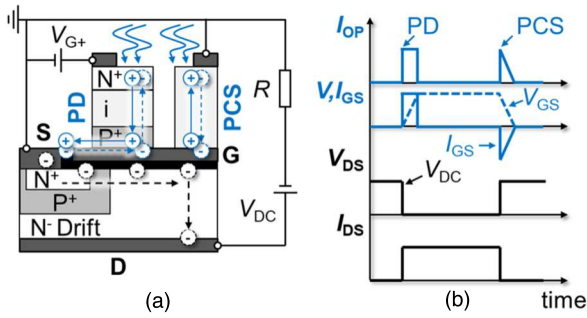


Fig. 8. (a) Device schematic and (b) ideal switching waveforms of a hybrid optical-gating MOSFET. The blue symbols represent photogenerated carriers, while the black symbols represent the carriers supporting the main device conduction current.

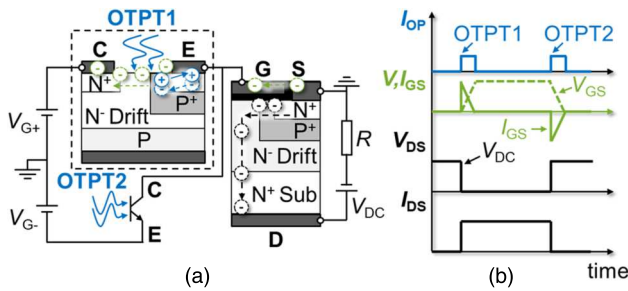


Fig. 9. (a) Device schematic and (b) ideal switching waveforms of an OTPT-based optically-driven SiC MOSFET. The blue symbols represent photogenerated carriers, the green symbols represent the carriers supporting the main device conduction current, and the purple symbols indicate gate-charging electrons.

Overcoming the complexity of customized OTPTs, the PD-based optical device represents a highly reliable and universally applicable strategy.^{108,109)} As shown in Fig. 10(a), this approach replaces the custom OTPTs with a totem-pole arrangement of two reverse-biased PDs. Note that, for driving WBG power devices, InGaAs PDs can be selected, which eliminate the need for UV optics.¹⁰⁸⁾ As shown in the ideal waveforms [Fig. 10(b)] and dynamic transient models [Figs. 10(c) and 10(d)], the two PDs are

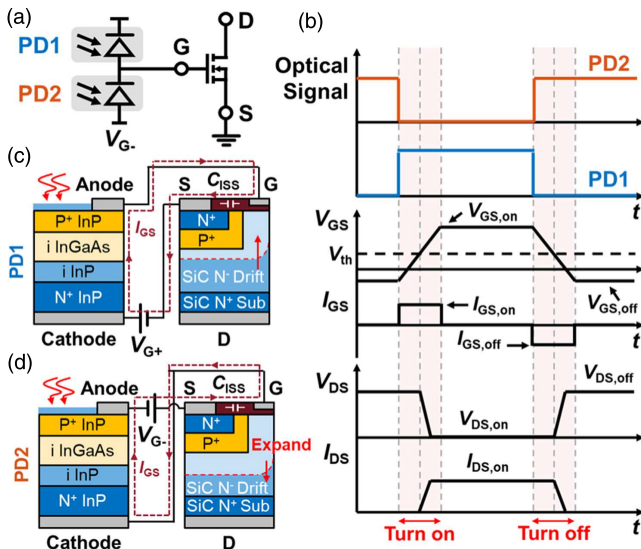


Fig. 10. (a) Schematic of a PD-based optically-driven SiC MOSFET and (b) ideal switching waveforms of the two PDs and the power device. Dynamic behavior of the device during the (c) turn-ON and (d) turn-OFF processes. Figures are reproduced from Fig. 1 in Ref. 108.

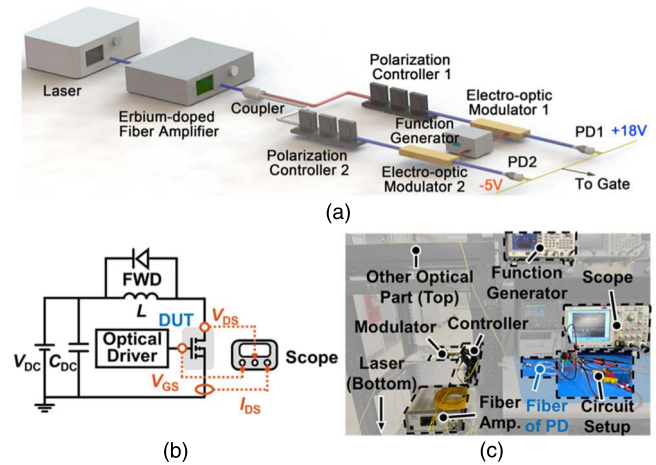


Fig. 11. (a) Schematic of the fiber-optic driving system. (b) Circuit schematic and (c) photograph of the DPT setup. Figures are reproduced from Fig. 2 in Ref. 108.

illuminated by complementary optical signals, effectively emulating an ideal current-source gate driver. Figure 11(a) shows the prototyped fiber-optics system used to drive this topology in Ref. 108, while the double-pulse test circuit and physical setup for a 3.3 kV SiC MOSFET are shown in Figs. 11(b) and 11(c). Utilizing a low power of 21.7 mW per PD, this configuration executed 1500 V/3 A hard-switching with rise and fall times of 152 ns and 215 ns, respectively, representing the fastest switching speeds and lowest optical energy requirements reported for high-voltage optically-controlled unipolar devices.¹⁰⁸⁾

This topology is equally effective for GaN architectures, as demonstrated by the rapid switching waveforms of a 650 V GaN HEMT shown in Fig. 12.¹⁰⁹⁾ Specifically, the device achieves an ultrafast fall time of 63.2 ns and a rise

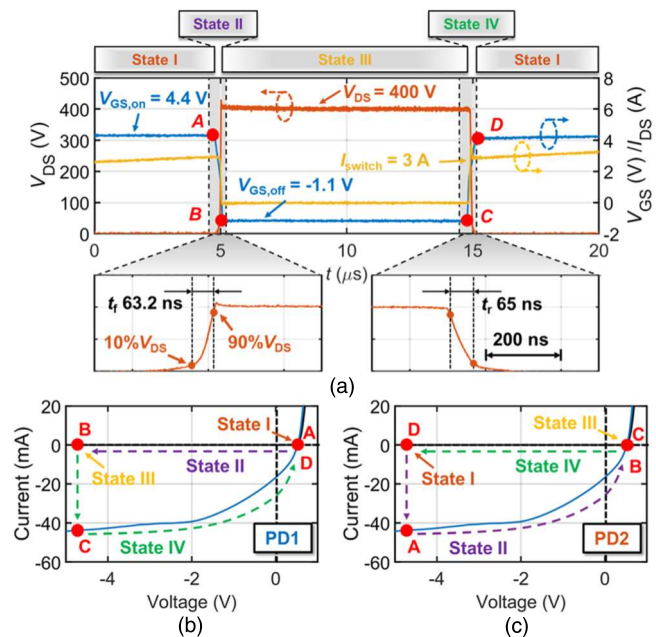


Fig. 12. (a) 400 V/3 A DPT waveforms of the GaN HEMT under 45 mW of peak P_{OP} on each PD. Switching logic of (b) PD1 and (c) PD2, with marked states and operation points corresponding to (a). The blue and black curves represent the PD's I - V characteristics under 45 mW P_{OP} and dark conditions, respectively. Figures are reproduced from Fig. 5 in Ref. 109.

Table II. Key parameters of optical-controlled power switches reported in the literature.

Device type		Switching power	Turn-on/off speed	Optical power
PCSS		4.25 kV/70 A(peak)	30 ns pulse	13 mJ in 30 ns
Thyristor		7.5 kV/4 kA(peak)	10 μ s pulse	40 mW
Direct Optical Gating	GaN FinFET	3 V/29 mA	$\sim 1/>10$ s	1 μ W
	GaN HEMT	30 V/11 mA	1.6/0.873 μ s	245 mW
Hybrid Optical Gating	Si MOSFET	12 V/20.7 A	$\sim 50/50$ ns	50 W cm ⁻²
OTPT Optical Driving	SiC MOSFET	500 V/0.77 A	100/200 ns	4 W
PD Optical Driving	SiC MOSFET	1.5 kV/5.6 A	156/160 ns	42 mW
	GaN HEMT	400 V/6 A	65/63.2 ns	90 mW
	GaN JFET	400 V/6 A	990/915 ns	90 mW

time of 65 ns during 400 V/3 A hard-switching transitions. Furthermore, the figure shows the unique switching locus of the photodiodes (PD1 and PD2) during operation. Unlike typical optoelectronic applications where PDs operate in a relatively static sensing bias, the PDs in this driver experience large-signal, dynamic transitions between the photoconductive mode (under reverse bias) and the PV mode (under forward bias) to effectively charge and discharge the gate. Proving its broad versatility and universality, this PD-based driving scheme has also been successfully demonstrated on current-driven devices, such as the 1200 V vertical GaN JFET.¹¹⁰⁾ Table II summarizes key parameters for these optical devices, including device types, switching power, turn-on/off speed, and applied optical power.^{37,90,93,98,104,108,109)}

While the illumination for signal-level receiver with GHz-level bandwidths is typically uniform (e.g. InGaAs PDs with microscopic active areas ranging from tens of micrometers to sub-millimeters¹¹¹⁾), it becomes a critical electro-optical challenge as optically controlled switches scale to handle power-level currents. If a large-area power chip is illuminated by a highly focused or Gaussian optical beam, the active regions will switch asynchronously. During fast turn-off operations, this forces the load current to crowd into specific regions, generating severe localized hotspots that can lead to catastrophic failure—a phenomenon well-documented in the historical development of large-area LTTs.^{105,112)} Although current crowding is not yet the primary failure mode in currently prototyped, relatively small WBG devices, it will inevitably become a fundamental bottleneck for high-current scaling. As demonstrated in OTPT designs,^{38,105)} suppressing this effect requires advanced optical delivery solutions that provide a spatially uniform optical wavefront rather than a narrowly focused beam. Irradiating the large-area device uniformly is essential to eliminate lateral propagation delays and mitigate nonuniform current densities during switching transients.

Finally, instead of completely relying photonics for gate driving, optical triggering has also been recently proposed to aid the electrical driving, such as the synchronized photonic-electronic driving reported for p-GaN gate HEMTs.^{113–116)} Synchronous UV illumination supplements the channel's photo-electric conductivity during turn-on, reducing the required electronic gate overdrive by over 2 V.^{114,115)} During turn-off, the active electrical gate rapidly sweeps out photo-induced carriers to mitigate the persistent photo-current tail and stabilize the threshold voltage.^{113,116)}

4. Optocoupler-based signal isolators: performance and reliability

4.1. Signal isolation types

In modern power electronics, highly reliable signal isolators are strictly required to enforce galvanic separation between the signal and power domains. These isolators play a pivotal role in delivering precision pulse-width modulation (PWM) signals to gate drivers, isolating feedback control loops, and transmitting fault detection signals without establishing conductive ground loops.

To achieve this critical barrier, the industry relies on three primary isolation technologies: magnetic isolation (transformer-based), capacitive isolation, and optical isolation (optocoupler-based), as shown in Fig. 13. Magnetic isolators utilize inductively coupled coils across a polyimide or silicon dioxide (SiO₂) barrier. Recent advancements in coreless PCB transformers have demonstrated high-frequency performance, achieving coupling capacitances as low as 1.9–5.85 pF and supporting common-mode transient immunity (CMTI) exceeding 100 V ns⁻¹ for 10 kV SiC applications.^{34,117)} Capacitive isolators similarly utilize an internal SiO₂ dielectric but rely on modulating a high-frequency carrier signal across the capacitive barrier. While both magnetic and capacitive isolators offer high-speed data transfer and high-density integration, they fundamentally rely on transient electric or magnetic fields, making them susceptible to specific high-frequency noise injection.

In contrast, optical isolation achieves electrical decoupling by converting input signals into photons via an LED and reverting them to the electrical domain through a photo-detector. This transmission occurs across a transparent, highly insulating silicone or glass dielectric gap. Because signal transfer is strictly photon-based, yielding high DC

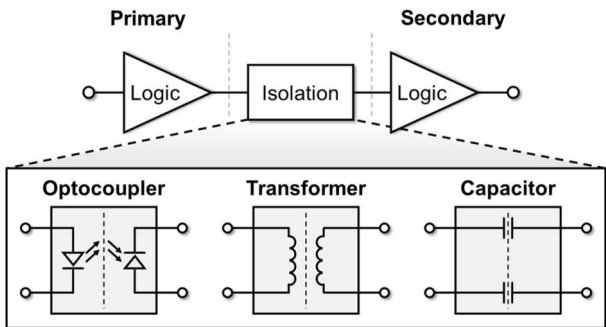


Fig. 13. Common isolation technologies, including optocoupler-, transformer-, and capacitor-based barriers.

Table III. Comparison of key parameters for isolated gate drivers.

Part number	Isolation type	Isolation voltage (kVrms)	Isolation capacitance (pF)	Size (mm ²)	Delay (ns)	CMTI ($\pm$ V/ns)	Publish year
TLP5814H	Optocoupler	5	1	58.5	150	70	2025
ACPL-W346		5	0.6	52.7	120	100	2018
1ED314xMU12F	Transformer	5.7	1	51.5	40	300	2025
2EDF7275F		5.7	2	60	37	150	2025
UCC21755	Capacitor	5.7	1	111	130	150	2024
STGAP3S		5.7	1	100	75	200	2024

resistance and minimal parasitic capacitance, optocouplers are inherently immune to extreme EMI. This makes them suitable for robust gate driving protection in harsh environments, like high-voltage Marx circuits for vehicle-integrated systems.^{118,119} Table III shows a comparison of the state-of-the-art industrial isolators, detailing their isolation types, isolation voltage ratings (kVrms), parasitic isolation capacitance (C_{IO}), physical footprints, propagation delays, and CMTI limits, is summarized in Refs. 120–125.

4.2. Reliability of optocoupler

Despite EMI immunity, current commercial optocouplers often face reliability challenges that should be considered for power electronics system design.¹²⁶ The most critical operational metric for an optocoupler is the current transfer ratio (CTR), defined as the ratio of the output photodetector current to the input LED drive current. The primary degradation concern in optocouplers is the gradual degradation of this CTR over its operational lifespan.^{127,128} As shown in Fig. 14(a), accelerated degradation tests are commonly employed to model the lifetime estimation of optocouplers. The underlying physics of failure predominantly point to the degradation of the LED emitter rather than the photodetector.¹²⁹ Over prolonged operation, thermal and electrical stresses induce crystal dislocations and defect migration within the highly forward-biased p-n junction of the LED. These defects act as non-radiative recombination centers, actively depressing the quantum efficiency of the light emission and subsequently causing the CTR to plummet unpredictably.¹³⁰ To further contextualize optical device endurance, Fig. 14(b) shows the characteristic lifetime degradation curves of high-power GaAs-based semiconductor diode lasers, showing how structural strain and operational temperature massively dictate the time-to-failure in III–V photonic devices.¹³¹

Temperature dependency is another pivotal factor in evaluating optocoupler performance, particularly because future high-density power modules are expected to operate

in excess of 200 °C.¹³² The thermal characteristics of optical isolators are intricately tied to the fundamental semiconductor physics of the constituent LED and photodetector. The reported variation of CTR as a function of ambient temperature (T) is presented in Fig. 15(a), showing a distinct roll-off at elevated temperatures.¹³² Furthermore, Figs. 15(b) and 15(c) show the reported temperature-dependent propagation delay and leakage current, respectively.¹³³ While advanced low-temperature co-fired ceramic (LTCC) packaged optocouplers exhibit highly stable delay times up to 250 °C, the leakage current inside the PD naturally increases exponentially due to the thermal generation of charge carriers.¹³³ As temperature increases, the thermal expansion of the crystal lattice and enhanced electron-phonon scattering cause the semiconductor bandgap to narrow, leading to a measurable redshift in the emission spectrum. As a representative example of this phenomenon, the temperature-dependent shift of the peak emission wavelength for typical green and blue LEDs reported by Ref. 134 is captured in Fig. 15(d).

Beyond the intrinsic semiconductor die, the physical packaging of the optocoupler could introduce thermo-

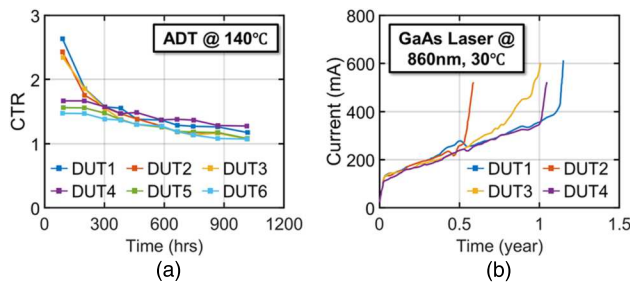


Fig. 14. (a) ADT results of six optocouplers at 140 °C (data points are reconstructed from Fig. 5 in Ref. 129) and (b) life-test characteristics of GaAs lasers operating at 0.86 μ m (data points are reconstructed from Fig. 1 in Ref. 131).

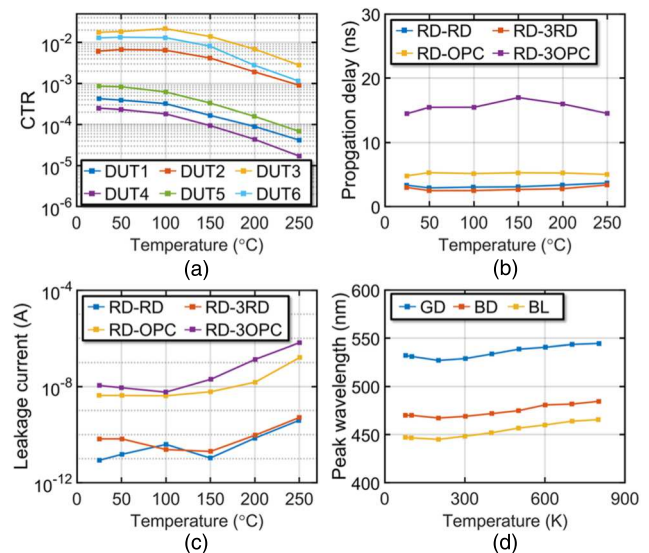


Fig. 15. (a) CTR of six AlGaInP optocouplers at varying temperatures (data points are reconstructed from Fig. 4 in Ref. 132). (b) Propagation delay (data points are reconstructed from Fig. 10 in Ref. 133) and (c) leakage current of four types of optocouplers at various temperatures (data points are reconstructed from Fig. 8 in Ref. 133). The DUTs are named based on their emitter/detector configurations: Red for Display/Red for Display (RD-RD), Red for Display/3 paralleled Red for Display (RD-3RD), Red for Display/OPC-6900-21 (RD-OPC), and Red for Display/3 paralleled OPC-6900-21 (RD-3OPC). (d) Peak positions of electroluminescence (EL) spectra as a function of temperature for three DUTs: green LED for display (GD), blue LED for display (BD), and blue LED for lighting (BL) (data points are reconstructed from Fig. 1 in Ref. 134).

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mechanical reliability concerns. Figure 16 shows the typical cross-sectional packaging structure of a standard plastic-encapsulated optocoupler.¹³⁵⁾ As the various constituent materials, including the copper lead frame, silicon die, gold wires, and plastic epoxy molding compound, possess vastly different coefficients of thermal expansion (CTE), severe temperature cycling within a power converter induces internal shear stresses. These stresses frequently culminate in wire-bond fatigue, ball-bond liftoff, or complete wire disconnection, immediately causing an open-circuit failure.¹³⁶⁾ In extreme cases, repeated thermal shock can propagate stress directly into the brittle semiconductor material, resulting in the catastrophic physical chip damage.¹³⁶⁾ Additionally, if the plastic encapsulation is non-hermetic, moisture and ionic contaminants can ingress along the lead-frame interfaces. Upon reaching the biased semiconductor die, this moisture initiates electrochemical migration and aluminum pad corrosion, causing severe damage to the internal contacts.^{135,136)} To mitigate these packaging failures in extreme environments, recent developments have shifted toward hermetically sealed LTCC platforms, which offer closely matched CTEs and superior thermal dissipation, enabling reliable optical galvanic isolation at operating temperatures pushing 250 °C.^{132,133)}

In summary, optocoupler-based signal isolators provide galvanic isolation and CMTI by transmitting data via photons, making them suitable for driving noise-sensitive WBG power devices. However, designers should consider the intrinsic, time-dependent degradation risks of the LED's quantum efficiency (CTR loss) and the complex thermo-mechanical vulnerabilities of standard plastic encapsulation. By leveraging predictive accelerated lifetime models and adopting robust, high-temperature packaging technologies like LTCC, the reliability of optical isolators can be extended to meet the rigorous demands of next-generation power electronics.

5. Photovoltaic cell-based auxiliary power supplies

5.1. Working principles of optical power supply

In high-power and MV converter systems, the APS is a mission-critical component responsible for energizing low-voltage loads such as gate drivers, digital controllers, and sensing amplifiers.²⁷⁾ The design of the APS becomes exceptionally demanding for high-side switching devices. Depending on the semiconductor technology and switching frequency, these gate drivers typically require a continuous power delivery ranging from a few watts up to several tens of watts.²⁷⁾ As shown in Fig. 17, the switching node of a high-side device experiences high-frequency dv/dt transients. To ensure the integrity of the gate driving voltage ($V_{\text{APS-G}}$) and prevent severe control failures, the APS provides reliable galvanic isolation between the low-voltage control domain and the high-voltage power stage.

Traditional isolated APS architectures, as shown in Fig. 18(a), rely on a multi-stage electrical conversion process: a DC input is inverted to high-frequency AC, transmitted across an isolation barrier using a magnetic-core transformer, and finally rectified back to a continuous DC output.^{27,137)} While recent advancements have utilized specialized inductor–capacitor–capacitor–inductor-LC resonant networks or diode–capacitor networks to push the

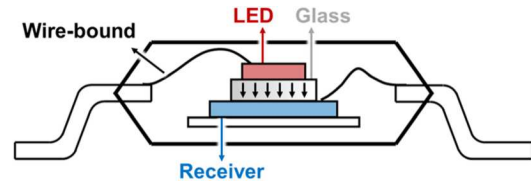


Fig. 16. Typical packaging structure of a plastic optocoupler.

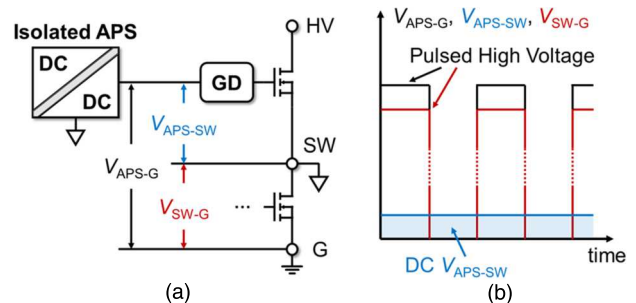


Fig. 17. (a) Schematic of a gate driving circuit for power devices, indicating the voltages from the APS to the switch node ($V_{\text{APS-SW}}$), APS to ground ($V_{\text{APS-G}}$), and switch node to ground ($V_{\text{SW-G}}$). (b) Ideal voltage waveforms of $V_{\text{APS-SW}}$, $V_{\text{APS-G}}$, and $V_{\text{SW-G}}$, illustrating the voltage jump of $V_{\text{SW-G}}$ between the high-voltage bus and zero, along with the continuous DC voltage for $V_{\text{APS-SW}}$.

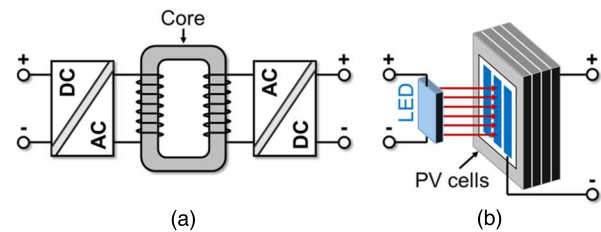


Fig. 18. Isolated auxiliary power supplies based on (a) magnetic coupling (i.e. transformers) and (b) PV cells.

partial-discharge inception voltages beyond 15 kV,^{138–140)} the physical proximity required for magnetic coupling intrinsically generates parasitic inter-winding capacitance. Under nanosecond switching transients, this parasitic capacitance acts as a direct conduit for common-mode EMI currents to flood the low-voltage control circuits. To circumvent these EMI bottlenecks, photonic power conversion offers a promising alternative [Fig. 18(b)]. In this approach, energy is transmitted via photons across a highly insulating optical barrier to a PV receiver. Because a single PV cell only generates a small forward voltage, multiple PV cells are monolithically stacked in series to aggregate their individual open-circuit voltages, achieving the desirable gate-driving voltage while maintaining near-zero coupling capacitance. By utilizing light to transmit power across a transparent insulating medium, the system achieves high isolation and near-zero coupling capacitance, making it inherently immune to EMI.^{141,142)}

The core energy-harvesting component of the optical APS is the PV cell. The fundamental electrical characteristics of these devices under monochromatic illumination are shown by the current–voltage (I – V) and power–voltage (P – V) curves in Fig. 19. At low operating voltages, the photo-generated current remains exceptionally steady and flat, acting as an ideal current source. However, as the applied forward bias approaches the built-in potential of the p–n

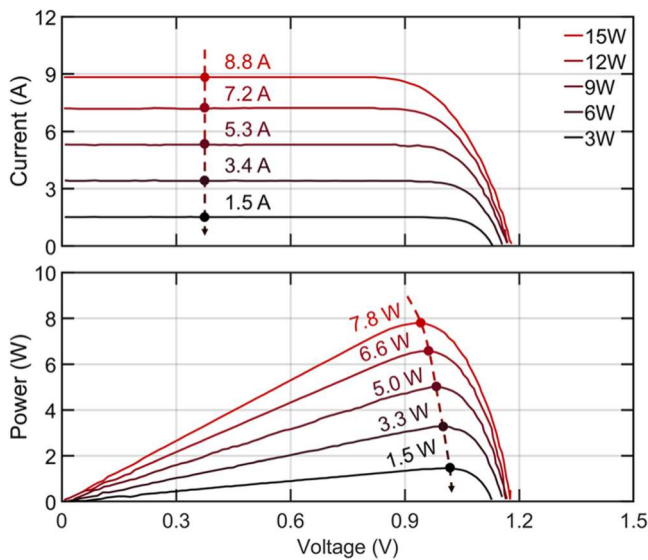


Fig. 19. Current-voltage (top) and power-voltage (bottom) characteristics of a GaAs-based PV cell illuminated by an 809 nm laser at different optical power levels ranging from 3 to 15 W. Data points are reconstructed from Fig. 5 in Ref. 143.

junction, the current drops precipitously to zero at the open-circuit voltage (V_{OC}). The magnitude of the steady-state photocurrent scales proportionally with the incident optical power. Consequently, for any given optical illumination level, there exists a specific maximum power point on the P - V curve that the downstream regulating circuit must track to maximize energy extraction.

For high-performance optically-isolated APSs, the desirable properties of a PV receiver include a high open-circuit voltage (V_{OC}) to minimize the required number of series-stacked cells, and a high optical-to-electrical conversion efficiency at the specific wavelength of the driving laser. The fundamental limits of these parameters are strictly dictated by the semiconductor material properties. As summarized in Fig. 20, the achievable V_{OC} of a reported PV cell increases near-linearly with the material's energy bandgap (E_g).¹⁴⁴⁾ WBG materials structurally suppress intrinsic carrier concentrations and thermal recombination currents, allowing the quasi-Fermi levels to split further apart under illumination, thereby yielding inherently higher operating voltages.

Furthermore, the theoretical optical-to-electrical conversion efficiency is wavelength-dependent. As shown in Fig. 21(a), the reported theoretical efficiencies for InGaP,

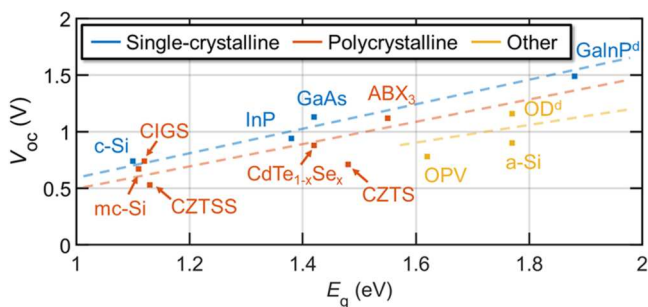


Fig. 20. Open-circuit voltage versus bandgap for PV cells based on different semiconductor materials. Data points are reconstructed from Table 1 in Ref. 144.

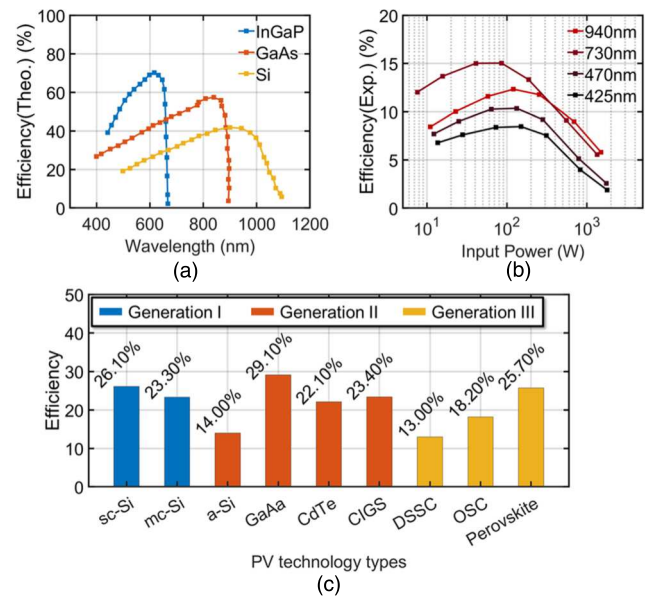


Fig. 21. (a) Theoretical efficiency versus wavelength for common PV materials (InGaP, GaAs, Si) (data points are reconstructed from Fig. 5 in Ref. 33). (b) Experimental efficiency as a function of input power for a single PV cell under various illumination wavelengths (data points are reconstructed from Fig. 6 in Ref. 145). (c) Maximum efficiencies achieved by different PV cell technologies (data points are reconstructed from Fig. 7 in Ref. 146).

GaAs, and Si cells peak sharply when the incident photon energy slightly exceeds the material's bandgap.³³⁾ This optimizes photon absorption while minimizing thermalization losses; at shorter wavelengths, excess energy is lost as heat, whereas longer wavelengths pass through the material entirely undetected. Beyond theoretical limits, experimental efficiency is also a strong function of illumination intensity. For instance, as reported in Ref. 145 and shown in Fig. 21(b), the conversion efficiency of a typical PV cell initially increases with optical input power due to the saturation of Shockley-Read-Hall recombination defect-mediated recombination. However, under extreme optical intensities, the efficiency inevitably rolls off due to severe series resistance losses and Auger recombination.¹⁴⁶⁾ Consequently, material selection is critical for maximizing system performance. While traditional crystalline silicon dominates broadband terrestrial solar harvesting, highly efficient single-crystal III-V materials, particularly GaAs, remain the gold standard for laser-driven photonic power conversion. As highlighted by the maximum efficiencies summarized in Fig. 21(c),¹⁴⁶⁾ these direct-bandgap materials offer monochromatic photon absorption, making them suited to pair with the narrow-emission spectra of commercial high-power laser diodes.

5.2. Demonstration and EMI improvement

When comparing isolation technologies, Fig. 22 highlights the distinct trade-offs in efficiency and volumetric footprint.¹⁴⁵⁾ While core-based magnetic supplies offer high end-to-end efficiency, their bulk and capacitive coupling limit their integration in ultra-compact, high-frequency modules. Photonic isolated supplies accept a modest reduction in total efficiency in exchange for a high degree of miniaturization and EMI immunity.³³⁾ In practical circuit implementations, two primary system architectures are utilized, as shown in Fig. 23. The first architecture

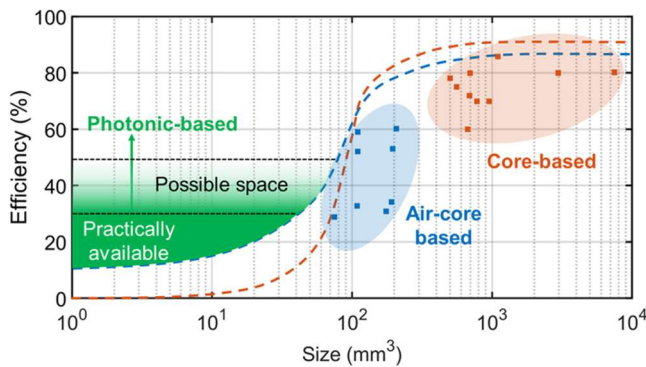


Fig. 22. A perspective on the efficiency and size of isolated auxiliary power supply technologies: air-core based, magnetic core-based, and optical power transfer, highlighting the theoretical and practically achievable opportunity space for optical power delivery. Data points are reconstructed from Fig. 2 in Ref. 145.

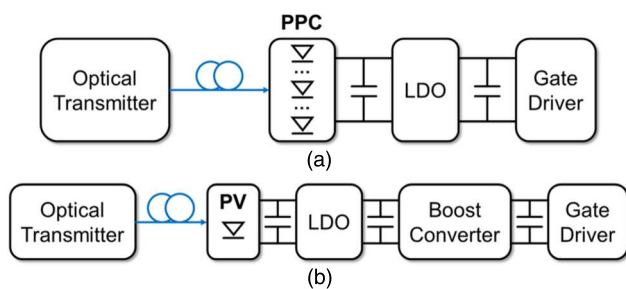


Fig. 23. Schematics of optical power delivery systems based on (a) a PPC and (b) a PV cell paired with a boost converter.

[Fig. 23(a)] employs a specialized photonic power converter (PPC), featuring multiple vertically stacked p-n junctions connected in series. This allows the PPC to directly output higher, usable DC voltages (e.g. 5, 10, 15, or 20 V), which are then smoothed by a low-dropout regulator to directly feed the gate driver.¹⁴²⁾ The second architecture [Fig. 23(b)] utilizes a standard single-junction PV cell, which typically generates less than 1 V. This low-voltage output must be fed into a local high-gain boost converter to step the voltage up to the required 15 V or 20 V driving levels.¹⁴⁷⁾

Hardware demonstrations have proven the viability of these photonic APS systems in high-voltage environments. Utilizing the single-cell architecture, Zhang et al. developed a power-over-fiber APS to drive a 15 kV SiC MOSFET.¹⁴⁷⁾ By delivering continuous laser power over a long, flexible optical fiber to a PV cell and boost converter, the system achieved over 20 kV of galvanic isolation while decoupling the control stage from the SiC device's dv/dt transients. Despite being optically powered, the circuit maintained robust protection functions, responding to overcurrent faults within a rapid 400 ns timeframe. Pushing the limits of the stacked-junction architecture, Ishigaki et al. demonstrated an advanced 12 V optical power transducer comprising 12 vertically stacked GaAs p-n junctions.¹⁴⁸⁾ By precisely current-matching the 12 sub-cells, this structure eliminates the need for a secondary switching boost converter, achieving a high open-circuit voltage of 14.5 V and an optical-to-electrical conversion efficiency of 66%. Driven by an 840 nm laser, the system delivered over 3.2 W of electrical power with less than 1 pF of stray capacitance, supporting ultra-fast $75 \text{ kV } \mu\text{s}^{-1}$ switching transients with an overall system efficiency of 25%.

The stability and noise-suppression capabilities of stacked PPCs are further highlighted in the work Wilkins et al.¹⁴²⁾ The hardware implementation is shown in Figs. 24(a)–24(c), including the laser transmission module and the miniaturized multi-junction PPC chip, which is capable of sustaining extreme areal power densities up to 100 W cm^{-2} , demonstrating a real need for the eventual integration of aggressive near-junction thermal management technologies.¹⁴⁹⁾ When implemented as a linear photonic boost-mode power supply stepping up a standard 3.3 V input to a regulated 12 V output [Fig. 25(a)], the PPC demonstrates a remarkable advantage in voltage regulation. As plotted in the time and frequency domains in Figs. 25(b) and 25(c), conventional electrical switching power supplies exhibit substantial high-frequency voltage ripples (over 20 mV), whereas the continuous-wave optical delivery of the PPC results in a virtually ripple-free DC output that is limited only by the measurement noise floor. During active converter operation driving a high-speed GaN transistor [Fig. 25(d)], the PPC-based gate driver easily matches the rapid, sub-10 nanosecond switching speeds of traditional transformer-based drivers, but vastly outperforms them in isolation due to its near-zero coupling capacitance. As shown in Figs. 25(e) and 25(f), the conducted EMI current propagating from the power stage back to the control circuitry is dramatically reduced. Specifically, the high-frequency common-mode EMI noise spikes are attenuated by more than 20 dB μA compared to the electrical transformer case, effectively dropping the conducted noise down close to the baseline background.

In summary, PV-based APSs offer a promising solution to the severe isolation and EMI challenges of modern WBG converters. By replacing magnetic coupling with optical power transmission, these systems eliminate parasitic capacitance to achieve galvanic isolation. Whether utilizing single-junction cells paired with boost converters or monolithic multi-junction PPCs, optical supplies deliver ripple-free voltage regulation and dramatic reductions in both physical footprint and common-mode noise, enabling robust gate driving for next-generation power electronics.

6. Optical data transmission, combination, and integration for optical driver IC

6.1. Data transmission through optical fibers

In modern power electronics systems, the reliable transmission of data across high-voltage boundaries is just as critical as the transmission of power. In this context, the high-frequency PWM control signals that dictate the switching states of the power devices are fundamentally a continuous stream of binary data.¹⁵⁰⁾ Beyond simple ON/OFF

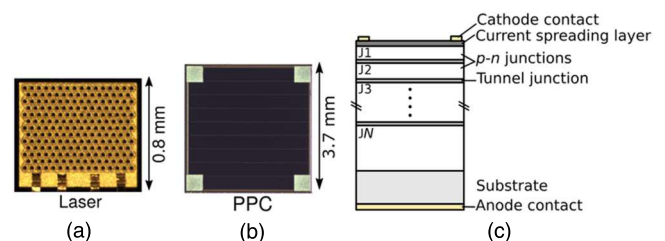


Fig. 24. Microscope images of (a) a laser diode chip and (b) a PPC. (c) Device schematic of the PPC showing its multi-junction layer structure. Figures are reproduced from Fig. 2 in Ref. 142.

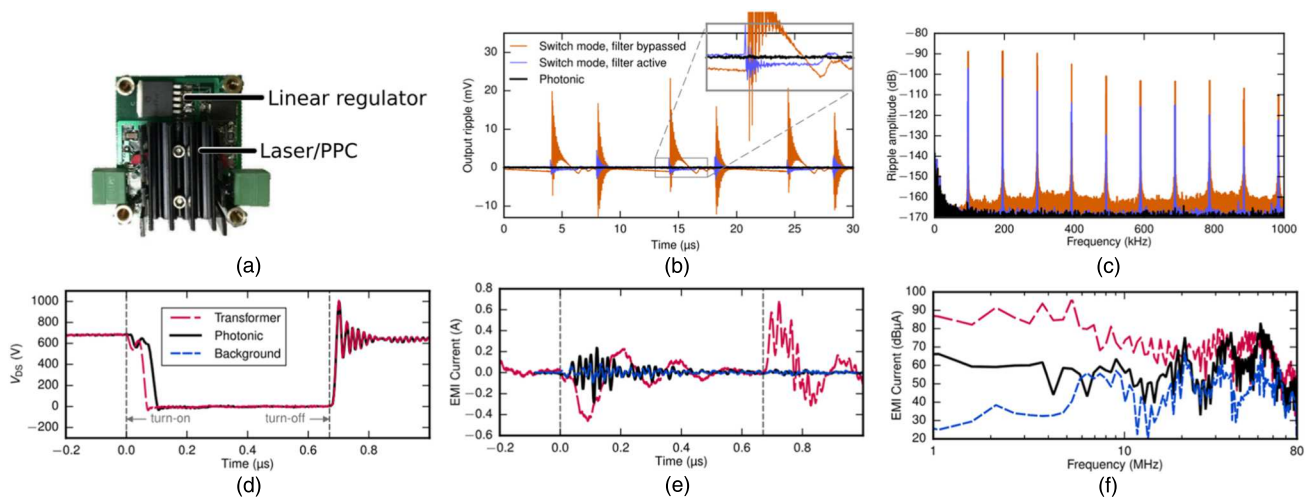


Fig. 25. (a) Photograph of an optical power supply based on a PPC (figure is reproduced from Fig. 5 in Ref. 142). (b) Voltage ripple and (c) frequency-domain representation of different auxiliary power supplies: a switching boost converter without a filter, a switching boost converter with a filter, and a PPC (data points are reconstructed from Fig. 5 in Ref. 142). (d) Switching waveforms of transformer-based and optically isolated auxiliary power supplies. (e) Measured time-domain and (f) frequency-domain EMI currents through the transformer-based and photonic power supplies (data points are reconstructed from Fig. 6 in Ref. 142).

commands, intelligent power modules increasingly require bidirectional data exchange to transmit telemetry, temperature readings, and fault codes (often requiring transmission rates of several bytes per microsecond) back to the central controller. Typically, high-speed primary switching commands are executed locally by an integrated microcontroller unit (MCU) or digital signal processor, allowing the optical link to function primarily as an auxiliary diagnostic bus. For such systems, standard industrial diagnostic protocols (e.g. Controller Area Network bus or RS-485) dictate a nominal bandwidth requirement between 100 kb s^{-1} and 1 Mb s^{-1} .¹⁵¹⁾ Furthermore, the electrical power consumed by this data transmission must remain negligible to prevent localized thermal bottlenecks in the driver circuit. Modern CMOS optical transceivers achieve highly efficient operation at $\sim 1\text{--}5 \text{ pJ bit}^{-1}$,^{152,153)} operating at these auxiliary data rates ensures the transceiver consumes only microwatts to a few milliwatts, contributing virtually no thermal stress. Moreover, the switching of WBG devices generates extreme dv/dt transients, this data transmission should occur with ultra-low propagation delays (typically in the nanosecond range) while maintaining immunity to common-mode noise.²⁸⁾

Consequently, data transmission through optical fibers has become widely adopted in advanced power electronics. Optical fibers provide reliable communication, galvanic data isolation, and inherent immunity to EMI.^{36,154)} A quintessential example of this architecture is the highly compact isolated gate driver developed by Rothmund et al. for 10 kV SiC MOSFETs, the schematic of which is shown in Fig. 26.¹⁵⁰⁾ In this system, fiber optic communication is utilized not only to deliver the high-speed PWM gate signals from the primary controller to the high-voltage side but also to establish a dedicated optical feedback loop for fast overcurrent protection. By separating the signal data from the electrical power domain, the system achieves a high 20 kV DC isolation rating and a low coupling capacitance of just 2.6 pF, protecting the SiC modules from short-circuit failures

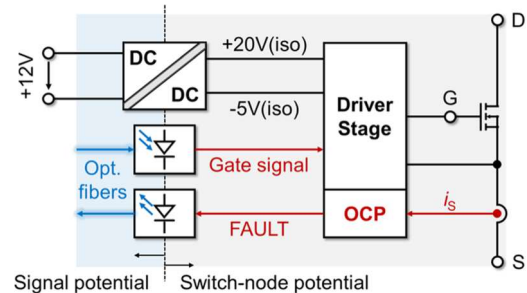


Fig. 26. Schematic of a power device control system based on bidirectional fiber-optic communication.

without introducing the EMI vulnerabilities typical of electrical feedback traces.

6.2. Integration of optical data transmission

While discrete optical isolation for data and auxiliary power has proven highly effective, the next evolutionary step for high-density power electronics is the combination of these optical functions. Multiplexing power delivery and data transmission through the exact same optical cable reduces the footprint, weight, and complexity of the system's isolation barrier.

To achieve combined power and data transmission, Heinig et al.¹⁵⁵⁾ developed a single-fiber link that utilizes amplitude-modulated light to simultaneously deliver gate-drive power and PWM data. At the receiver, a demodulation circuit extracts the control signal with an ultra-low latency ($< 500 \text{ ns}$), while a boost converter stpdf up the harvested optical energy to the 15–20 V required to drive SiC or Si devices. Alternatively, to address the inherent power limitations of optical delivery in pulse power applications, Fujita¹⁵⁶⁾ proposed a resonant gate-drive circuit controlled and powered entirely via optical fibers. This design incorporates an inductor that forms a parallel LC resonant circuit with the device's intrinsic input capacitance. By recycling the gate charge during switching, this resonant approach reduces the active power consumption, allowing a low-power optical supply to effectively sustain the operation of fast-switching, high-voltage devices.

As power converters become more intelligent, the data bandwidth requirements often exceed simple PWM signals. Addressing this, Helmers et al. demonstrated a 6 W optical power link integrated with high-speed bidirectional optical data transmission, designed to render large packets of telemetry data alongside gate signals.¹⁴³⁾ The complete system schematic is shown in Fig. 27. The transmitter unit provides regulated voltages to multiple localized modules, including a MCU, a thermoelectric cooler, and an optical modulator. The telemetry data is processed by the MCU, modulated via the laser driver, combined with the continuous optical power beam, and transmitted over the fiber. This combined architecture achieved a PV conversion efficiency exceeding 50%, delivering 6.2 W of continuous electrical power at 3.3 V and 5 V, while supporting stable downlink data transmission rates up to 115.2 kb s⁻¹.

To push the limits of combined optical transmission even further, Matsuura et al. utilized double-clad fiber (DCF) technology to simultaneously deliver over 40 W of electric power alongside high-speed optical data.¹⁵⁷⁾ The system schematic is depicted in Fig. 28. Unlike standard single-mode fibers that multiplex signals via time or wavelength, the DCF spatially separates the transmission mediums: the high-intensity optical power stream is guided through the wide inner cladding, while the sensitive optical data signals are confined to the central core. By coupling a customized, high-intensity PV power converter to the DCF output, the system can accept feed light power exceeding 20 W with an optical-to-electrical conversion efficiency of over 50%. This structural innovation mitigates nonlinear optical crosstalk and thermal damage, achieving a continuous electrical power delivery of 43.7 W over a single optical fiber, suited for intensely demanding remote antenna units or massive high-voltage converter stations.

6.3. Integration for optical driver IC

While discrete optical receivers and power converters provide excellent isolation, packaging these individual components on PCB invariably introduces parasitic inductances and capacitances via wire bonds and routing traces. To fully exploit the high-frequency capabilities of WBG devices, the next-level evolutionary step is the development of fully integrated optical driver integrated circuits (ICs). By leveraging mature CMOS technology, it is possible to monolithically integrate the photodetector (the optical receiver) directly alongside the complex analog signal processing circuits and the high-current gate buffer on a single silicon die. This monolithic approach reduces propagation delays, minimizes the overall package footprint, and ensures

superior CMTI by keeping the high-impedance optical sensing nodes entirely confined within the chip.

A monolithic optical driver ICs have been proposed and developed.^{158,159)} For example, Le et al. demonstrated a monolithic optical isolator utilizing a standard 0.35 μm CMOS process, with its schematic and microscope photograph shown in Fig. 29.¹⁵⁹⁾ This design integrates a PD with a transimpedance amplifier and subsequent voltage amplification stages, achieving a 10 MHz bandwidth when illuminated by a 450 nm laser. Taking this monolithic integration a step further, Fig. 30 shows an architecture where the optical receiver is embedded directly within the active area of a vertical power MOSFET.¹⁶⁰⁾ The embedded photoreceiver achieves a full width at half maximum bandwidth exceeding 300 MHz and a responsivity greater than 0.15 A W⁻¹ at 500 nm, effectively creating an integrated, optically-triggered power switch.

Moreover, optical signal reception and optical power delivery can both be integrated onto a single die, creating a fully autonomous gate drive solution. Figure 31 details the schematic and physical layout of a CMOS silicon-on-insulator (SOI) gate driver that embodies this concept.^{161,162)} This architecture utilizes a dual-optical receiver setup: one section acts as an on-chip PV cell for energy harvesting, while the other processes the high-speed PWM commands. Because the harvested optical voltage is typically low, a built-in switched-capacitor circuit and clock generator step the voltage up to the required gate driving levels. The SOI technology natively suppresses parasitic latch-up, ensuring high robustness under high dv/dt transients. Furthermore, this integrated platform supports digital optical communication. Utilizing a variable-frame-length serial protocol, the IC can receive a 4 bit commutation order in just 56 ns and an 8 bit configuration data packet in 84 ns (a data rate of 140 Mbps). This allows the central controller to use light not only to switch the WBG device but to reconfigure the internal gate resistor in real-time to actively shape the switching transients.¹⁶²⁾

7. Optical sensors for power electronics

In modern high-voltage grids and high-density power electronics, the continuous monitoring of voltage, current, and temperature is critical for system control and health management.¹⁶³⁾ Traditional electromagnetic instrument transformers are bulky, susceptible to EMI, prone to magnetic core saturation, and carry the risk of explosive failure.^{164–166)} Optical sensors overcome these physical barriers by providing intrinsic galvanic isolation. When light propagates through specialized crystals or optical fibers, external electric, magnetic, or thermal fields modulate the phase, wavelength, intensity, or polarization of the light. By analyzing these optical shifts, the system's states can be precisely quantified with high EMI immunity, wide bandwidth, and negligible risk of conductive ground loops. Given the commercialization and deployment of optical sensors across industrial and power grid applications,^{167,168)} this section will briefly introduce their fundamental working principles alongside a selection of representative works. Consequently, it is intended to serve as a foundational overview rather than a fully comprehensive survey of the commercial sensor landscape.

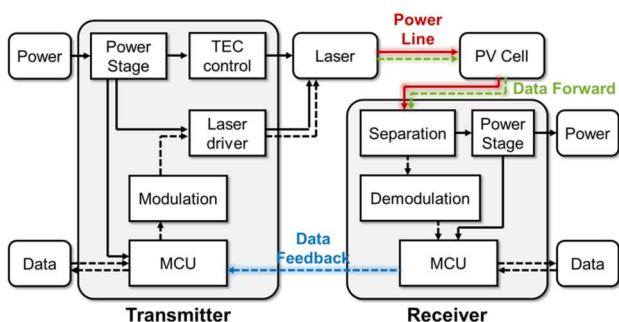


Fig. 27. System schematic of optical power delivery combined with optical data transmission.

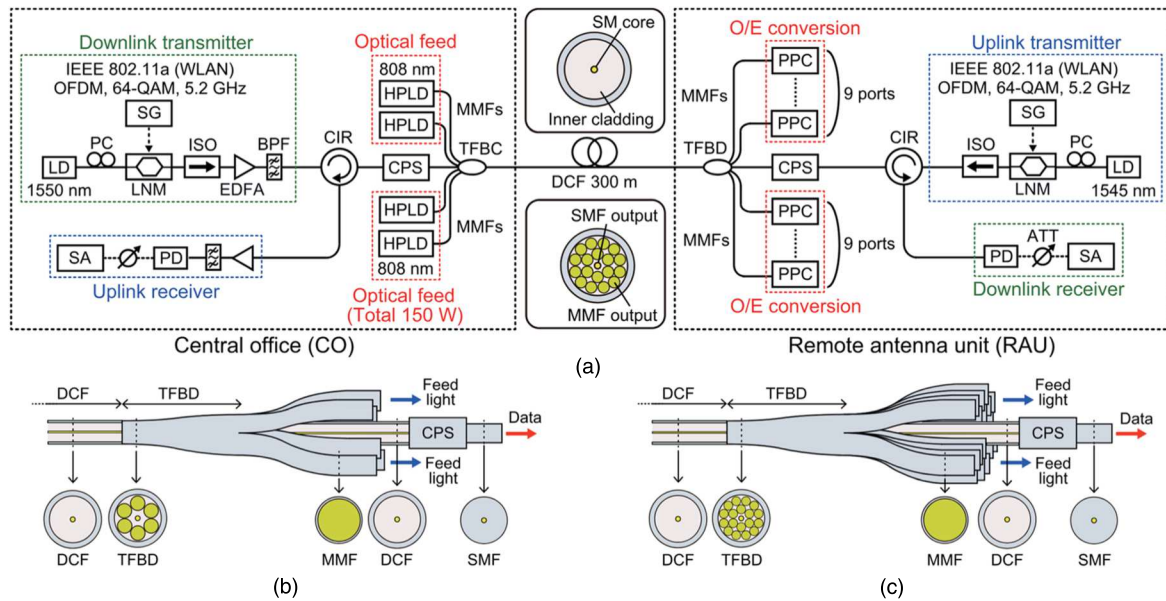


Fig. 28. (a) System schematic proposed in Fig. 2 in Ref. 157, consisting of a central office and a remote antenna unit, utilizing optical fibers to transmit both power and data. Configurations of a tapered fiber bundle divider with (b) six and (c) 18 multimode fiber output ports.

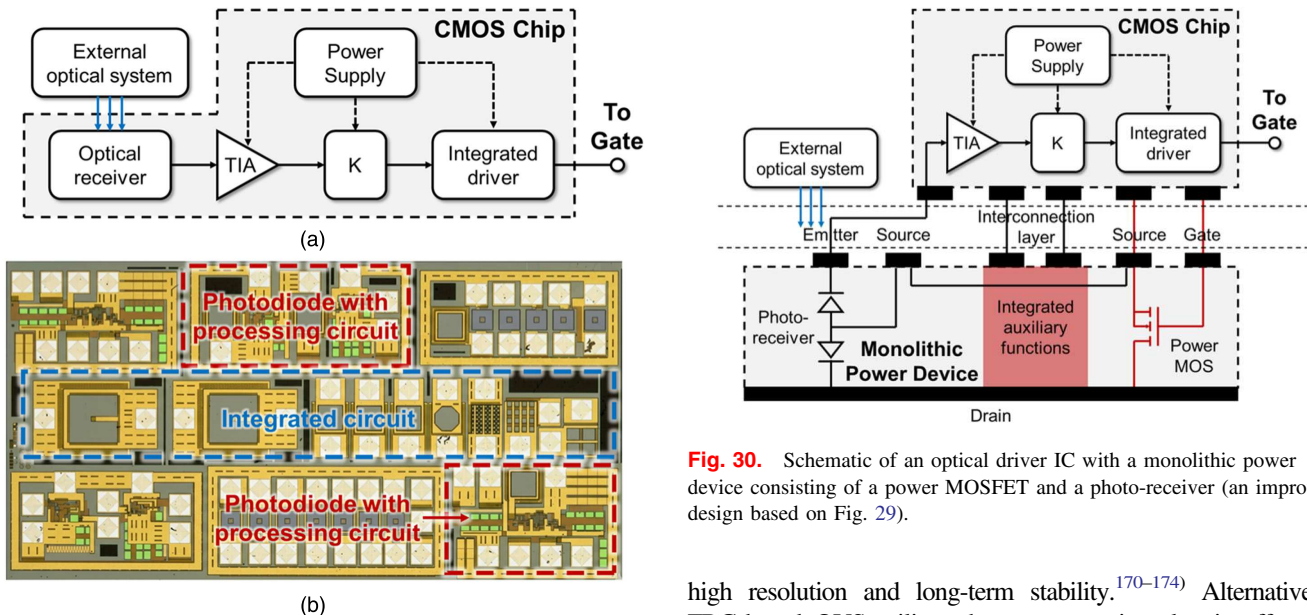


Fig. 29. (a) Schematic and (b) microscope photograph of an advanced optical gate driver IC, integrating an optical receiver, power supply, TIA, voltage amplifier with gain K, and a CMOS driver stage. (Figure is reproduced from Fig. 9 in Ref. 159.)

7.1. Optical voltage sensors

As shown in Fig. 32, a generalized optical voltage sensor (OVS) comprises an optical emitter, a high-voltage sensing element, and a receiver. Modern OVS technologies are primarily divided into electro-optic (EO) sensors and fiber Bragg grating (FBG)-based piezoelectric sensors, and microelectromechanical system (MEMS) electrostatic sensors.^{164,169} EO sensors rely on the Pockels effect in non-centrosymmetric crystals like lithium niobate (LiNbO_3), where an applied electric field induces birefringence and a measurable phase retardation.¹⁶⁴ To mitigate temperature-induced non-reciprocal errors and natural birefringence drifts, advanced architectures utilizing reciprocal dual-crystal compensation, radial polarization detection, and cascaded microwave photonic filters have been developed, achieving ultra-

Fig. 30. Schematic of an optical driver IC with a monolithic power device consisting of a power MOSFET and a photo-receiver (an improved design based on Fig. 29).

high resolution and long-term stability.^{170–174} Alternatively, FBG-based OVS utilizes the converse piezoelectric effect. A piezoelectric ceramic (e.g. PZT) deforms under an applied voltage, mechanically straining an attached FBG and causing a measurable shift in its reflected Bragg wavelength.^{169,175–178} Recent advancements in this domain have focused on advanced packaging to decouple thermal strain, the integration of optoelectronic oscillators for microwave-frequency readout, and the use of piezoelectric thin films for highly miniaturized AC voltage tracking up to 350 kVrms.¹⁷⁸ Furthermore, optical field and voltage sensing can be achieved by exploiting electrostatic forces acting upon MEMS architectures. In these configurations, an applied electric field generates electrostatic induction that physically deflects a micro-spring supported membrane or seismic mass; this microscopic displacement is then precisely measured using optical tracking techniques, enabling highly sensitive, distortion-free field measurements.^{179,180}

7.2. Optical current sensors

Optical current sensors (OCS) address the saturation and bandwidth limits of traditional current transformers by utilizing the Faraday magneto-optic effect. When linearly

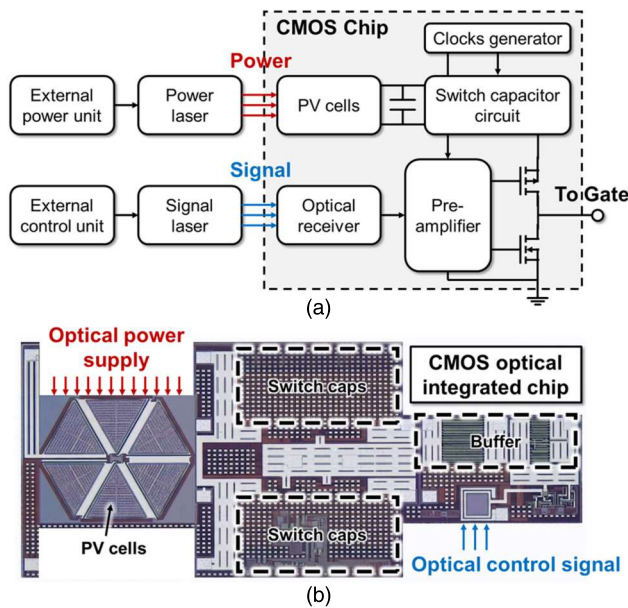


Fig. 31. (a) Schematic and (b) microscope photograph of an optical gate driver IC, combining optical power delivery and signal transmission. (Figure is reproduced from Fig. 2 in Ref. 161.)

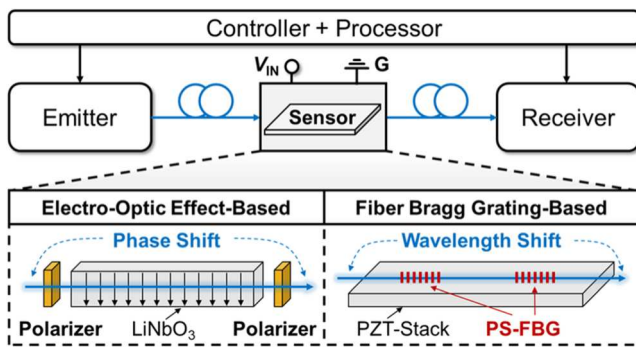


Fig. 32. Schematic of an optical voltage measurement system and the working principles of electro-optic effect-based and FBG-based optical voltage sensors.

polarized light passes through a magneto-optic medium exposed to a magnetic field, its plane of polarization rotates proportionally to the field intensity.^{167,181)} While bulk-glass OCS have been effectively deployed, their accuracy can be sensitive to conductor positioning and external stray magnetic fields.¹⁸¹⁾ Consequently, all-fiber OCS (FOCS) are widely preferred. By looping the sensing fiber around the conductor, FOCS perform a closed-path line integral of the magnetic field. According to Ampere's Law, this renders the sensor immune to external magnetic crosstalk.¹⁶⁷⁾ As shown in Fig. 33, advanced FOCS utilize polarization-rotated reflection interferometers to cancel reciprocal environmental noise. These robust, flexible architectures have successfully measured massive direct currents up to 500 kA with $\pm 0.1\%$ accuracy and allow for lightweight, "rope-shaped" deployment directly on live high-voltage lines.^{166,167,182)} Beyond conventional fiber-based Faraday rotation, emerging OCS technologies leverage magnetostrictive effects and advanced integrated magneto-optic architectures to achieve extreme miniaturization and wideband measurement. Magnetostrictive current sensors often utilize surface acoustic wave (SAW) devices coated with a magnetic film

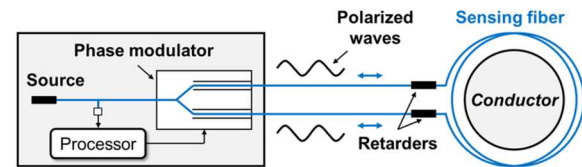


Fig. 33. Working principles and schematic of a fiber-optic current sensor.

(such as FeNi); the magnetic field generated by the target current induces mechanical strain in the coating, which alters the SAW propagation velocity and results in a measurable shift in the resonant frequency.¹⁸³⁾ Furthermore, to monitor highly transient, broadband currents, integrated optical waveguide sensors utilizing magneto-optical thin films have been developed. By coupling a magneto-optical thin film with a double-ring assisted Mach-Zehnder interferometer, these highly integrated devices can exploit magneto-optical nonreciprocity to achieve massive dynamic measurement ranges and ultra-high-frequency tracking capabilities up to the megahertz (e.g. 13.56 MHz) regime.¹⁸⁴⁾

7.3. Optical temperature sensors

Optical temperature sensors (OTS) leverage the temperature-dependent optical properties of dielectric materials, enabling non-contact, EMI-immune thermal profiling directly on high-voltage chips or within transformer oil.^{185,186)} FBG sensors are the most widely deployed technology for harsh environments. Thermal expansion and the thermo-optic effect cause a direct, linear shift in the FBG's reflected Bragg wavelength.¹⁸⁷⁾ Their miniature, chemically inert form factor is ideal for embedding within transformer oil-paper insulation, while specialized sapphire fibers enable distributed sensing at extreme temperatures up to 1300 °C.¹⁸⁵⁾ Alternatively, fluorescence-based OTS relies on rare-Earth-doped materials (e.g. Er^{3+} , Sm^{3+}) excited by a laser. Because the measurement relies on the time-domain decay lifetime of the fluorescence rather than absolute intensity, this technique provides highly precise thermal tracking inherently immune to optical power fluctuations.¹⁸⁶⁾ At the micro-scale, the strong thermo-optic effect of silicon can be leveraged in photonic integrated circuits (PICs), enabling dense, ultra-low-power on-chip temperature monitoring for future smart power modules.¹⁸⁸⁾

8. Emerging applications in quantum sensing and photonic systems

Quantum technology and its integration with photonics is receiving tremendous interest due to identified capabilities that exceed classical systems. These capabilities include enhanced sensitivity for sensors, more robust security for communications, and novel ways to transmit information in compact system architectures.¹⁸⁹⁾ In the context of this review, photonics-integrated power electronics have important ramifications for future technological areas involving quantum sensing and communications as follows.

Compact packaging of future quantum sensors involves the integration of power delivery, microwave signal delivery, photonics, control signal processing, and readout circuitry into a very small cm-to-mm scale form factor or package.^{190,191)} Here, electrical integrated circuits (EICs) must eventually be co-packaged with PICs. These EICs may

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include power converters, albeit at much lower output voltages than those found in traction drive mobility-related applications. In optically detected magnetic resonance-based sensing platforms, conventional power architectures may introduce supply-related noise or ripple that degrades laser driver and/or photodetector readout performance. For field-deployable sensors, these disturbances may lower sensitivity and decrease repeatability. To realize the full potential of quantum sensors, clean power delivery without noise and EMI is therefore critical to prevent coupling into sensitive measurement pathways. Thus, photonics-integrated power electronics and associated co-packaging technologies can further advance low-noise operation for tightly integrated quantum sensors where EICs and PICs must co-exist.

Building from the discussion above, it is expected that clean power delivery will also be essential for future photonics-integrated quantum network systems.^{189,192)} Such communications systems will logically involve future data centers connected to fiber networks that are fed by light sources which are then, in turn, connected to edge-deployed PIC platforms.¹⁹²⁾ It is a straightforward extension of this network architecture to then further employ EIC-PIC integrated quantum sensors at the edge to interact with future quantum communications network nodes. Hence, for heterogeneously integrated systems involving sensing, secure communications, and control functionality, power delivery that is free of EMI is again vital. Analogous to the development of advanced power supplies for today's data centers,¹⁹³⁾ many of the advanced photonics-integrated power conversion technologies reviewed herein are likely relevant for deploying high-fidelity quantum network technologies in the future.

9. Conclusion and outlook

The relentless pursuit of higher efficiency, switching frequencies, and power densities has pushed conventional electrical isolation technologies to their fundamental physical limits. This is particularly evident in the severe EMI challenges facing modern power systems, primarily caused by the rapid switching of WBG and UWBG power semiconductors. As comprehensively reviewed in this paper, photonics-integrated power electronics provides a promising solution to these fundamental bottlenecks. By replacing electrons with photons for control signals, auxiliary power, and data transfer, design engineers can eliminate galvanic pathways, achieving near-zero coupling capacitance and EMI immunity.

This review has systematically detailed the working principles, structural innovations, and state-of-the-art hardware demonstrations, with the key takeaways including:

- 1) The inherent ability of photonic links to transmit energy and information using photons across non-conductive, dielectric mediums eliminates capacitive coupling and provides immunity to system noise.
- 2) Several technologies, including optically-controlled unipolar power semiconductor switches, high-speed optical data links, and multi-junction PV power converters, play critical roles in constructing the photonics-based power electronics systems that deploy WBG and UWBG device technologies.
- 3) Photonic integration serves as an adaptable framework for power converters, spanning EMI-immune sensor

telemetry, PD-based gate driving, isolated PV-based auxiliary power, and quantum thread. On the other hand, achieving full functionality in photonics-based power electronics necessitates the synergistic integration of these diverse modules to ensure system-wide compatibility and performance.

Moving forward, transitioning this field from lab demonstrations into reliable industrial systems is expected to drive several exciting research directions:

- 1) Efforts are expected to explore advanced heterogeneous and monolithic integration, specifically the co-packaging or wafer-level bonding of photonic chips directly onto WBG and UWBG power dies. Such an integration is key to reducing propagation delays, shrinking module footprints, and lowering manufacturing costs.
- 2) Future research could greatly benefit from exploring high-temperature-resilient photonic materials, robust optical adhesives, and innovative thermal packaging architectures to ensure the long-term reliability of the optical isolation barrier.
- 3) Optimizing optical-gating architectures emerges as a highly promising strategy for WBG and UWBG optically-triggered unipolar power devices. Recent hardware demonstrations indicate that this scheme possesses the unique capability to concurrently achieve the ultra-fast switching speeds and low optical power consumption.
- 4) Expanding these direct optically-controlled power devices to handle MW-class power levels represents another compelling direction. This transition will likely involve a deeper exploration of material physics, device architecture innovations, and packaging to realize MW-class optically-triggered switches with kHz-level operation.
- 5) For PV cells utilized as APSs, enhancing the optical-to-electrical conversion efficiency remains a central challenge. Research may increasingly focus on multi-junction PV cells tailored for monochromatic high-power laser illumination, aiming to minimize thermalization losses and maximize overall system efficiency.
- 6) A promising long-term objective is the integration of high-speed bidirectional optical communication into a single link. This architecture could enable a single fiber to deliver uncorrupted, high-resolution telemetry data to support AI-driven condition monitoring and digital-twin modeling in next-generation power electronics system.

In conclusion, bridging the gap between photonics and power electronics represents a significant advancement for the field. By continuously innovating across device physics, material science, and advanced packaging, researchers will unlock the full potential of WBG systems, paving the way for a more resilient, highly intelligent, and highly EMI-immune electrified future.

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ORCID iDs

Xin Yang  <https://orcid.org/0009-0000-4036-2733>Yuhao Zhang  <https://orcid.org/0000-0001-6350-4861>

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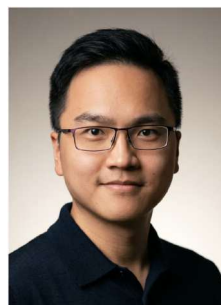
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Electronics Systems (CPES), Virginia Tech. His current research interests include the characterization, robustness, and reliability of wide-bandgap power devices.



University of Hong Kong, Hong Kong SAR, China. From 2023 to 2025, He was a Graduate Research Assistant with the Center for Power Electronics Systems (CPES), Virginia Tech, USA. From 2019 to 2020, he was with Fuji Electric at its Matsumoto factory, Japan, working on the reliability evaluation of SiC power modules. His research interests include the fabrication and characterization of wide-bandgap power devices.



Jizhong Jiang received the B.S. degree in 2020 and the M.S. degree in 2023 from the College of Electronic Science and Engineering, Jilin University, Changchun, China. He is currently pursuing the Ph.D. degree with the Center for Advanced Semiconductors and Integrated Circuits, and the Department of Computer and Electronic Engineering at The University of Hong Kong, Hong Kong. His research interests focus on high-performance optoelectronic and electronic devices, along with their practical applications.



Hengbo Zhang received the B.S. degree in Electrical Engineering and Automation from Sichuan University, Sichuan, China, in 2019, and the M.S. degree in Material Physics and Chemistry from University of Chinese Academy of Sciences, Beijing, China, in 2022. He is working toward the Ph.D. degree with the Department of Electrical and Electronic Engineering, the University of Hong Kong, Hong Kong SAR, China. His current research interests include the fabrication, characterization, and reliability of wide-bandgap power devices.

Feng Zhou, photograph and biography not available at the time of publication.

Shailesh N. Joshi, photograph and biography not available at the time of publication.



Ercan (Eric) M. Dede received his B.S. degree and Ph.D. in mechanical engineering from the University of Michigan and an M.S. degree in mechanical engineering from Stanford University. Currently, he is the Director of the Electronics Research Department at the Toyota Research Institute of North America. He is a Fellow of the American Society of Mechanical Engineers (ASME) and a Senior Member of the Institute of Electrical and Electronics Engineers

(IEEE). His team focuses on vehicle systems involving advanced sensors, human-machine interfaces, power semiconductors, electronics and photonics packaging, and thermal management technology. He has 260+ issued patents and has published more than 140 articles in archival journals and conference proceedings on topics related to advanced electronics systems including the design and optimization of thermal-fluid, mechanical, and electromagnetic devices. He is an author of a book entitled “Multiphysics Simulation: Electromechanical System Applications and Optimization.” His team has received two R&D 100 Awards for the development of technologies related to next-generation electronics for electrified vehicles, and he was co-recipient of the 2025 Harvey Rosten Award. He currently serves in an editorial role for IEEE Transactions on Components, Packaging and Manufacturing Technology.



Liyang Jin received the B.S. degree in electrical engineering and the M.S. degree in optical engineering from Chongqing University in 2020 and 2024, respectively. She is currently pursuing the Ph.D. degree in Electrical and Computer Engineering at Virginia Tech, Blacksburg, VA, USA. Her research interests include optical gate drivers, surface acoustic wave (SAW) gate drivers, GaN power electronics, and integrated photonics for power electronics applications.

Chao Xiang, photograph and biography not available at the time of publication.



Han Wang (Fellow, IEEE) received the B.A. degree in electrical and information science from Cambridge University in 2007 and his Ph.D. degree from Massachusetts Institute of Technology (MIT) in 2013. He is Professor in the Department of Electrical and Computer

Engineering and Director of Center for Advanced Semiconductors and Integrated Circuits (CASIC) at the University of Hong Kong. He is a Fellow of the Institute of Electrical and Electronics Engineers (IEEE). From 2013 to 2014, he was with the Nanoscale Science and Technology group at IBM T. J. Watson Research Center. In 2014–2023, he was with the Department of Electrical and Computer Engineering at the University of Southern California, first as an Assistant Professor and then as a tenured Associate Professor. In 2021–2023, he served as the head of the low dimensional materials research department at Taiwan Semiconductor Manufacturing Company (TSMC), concurrent to his faculty appointment. His research interests include microelectronics technology based on emerging semiconductor materials and devices for advanced logic, memory and AI hardware.



Linbo Shao is an Assistant Professor in the Bradley Department of Electrical and Computer Engineering at Virginia Tech. He was a Postdoctoral Fellow in the John A. Paulson School of Engineering and Applied Sciences at Harvard University from 2019 to 2021. He received his Ph.D. in Engineering Science in 2019 and M.S. in Applied Physics in 2016 from Harvard university, and B.S. in microelectronics in 2014 from Peking University. His current research interests include integrated microwave acoustic-wave

and optical devices and circuits for signal processing sensing, power electronics, computing, and quantum applications.



Yuhao Zhang is a Full Professor of Electrical and Computer Engineering at the University of Hong Kong. He received his B. S. in Physics from Peking University and his M.S. and Ph.D. in Electrical Engineering from MIT. Previously, he was the Shirish S. Sathaye Associate Professor at Virginia Tech. With over 200 publications and 9 granted U.S. patents, his research focuses on (ultra-)wide-bandgap power semiconductors and power electronics. He serves as Chair of the IEEE Electron Device Society Power

Devices and ICs Committee and Associate Editor-in-Chief for IEEE Electron Device Letters. His honors include the NSF CAREER Award, ONR Young Investigator Award, two IEEE George Smith Awards, four IEDM Technical Highlights, and the Compound Semiconductor Week Young Scientist Award.